Sequential Band Fusion for Hyperspectral Target Detection

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***Abstract*— Due to the enormous data and redundant information, how to process hyperspectral images reasonably and efficiently has become a research focus. This article proposes a sequential band fusion (SBF) approach for hyperspectral target detection and gives a detailed derivation of the fusion theory. Then four fusion algorithms—SBF based on band sequence (SBF-BSQ), SBF driven by an initial band (SBF-IBD), SBF based on band priority (SBF-BP), and SBF based on band selection (SBF-BS)—are introduced for application. Experimental results prove that the method proposed in this article can not only effectively improve the efficiency of target detection but also provide the trend of the detection value during the fusion process. Band fusion breaks through the limitation of band selection in the application and provides a new processing method for hyperspectral data.**

***Index Terms*— Band fusion, band sequential, sequential band fusion (SBF), target detection.**

I. Introduction

With the continuous development of remote sensing technology, the hyperspectral image tends to contain more spectral information in recent years. At the same time, there is a lot of redundant information in the image. Band selection (BS) is a method commonly used for data dimensionality reduction. However, there are a few problems with BS. First, the BS needs to converge effectively in a limited time to improve the efficiency of the algorithm. Second, the BS needs to ensure the performance of the optimal solution, that is, whether the attributes such as the information of the optimal band combination and the correlation between the bands have advantages after convergence. It is difficult for the BS to meet the requirements of low time-consuming and high performance at the same time. Supporting either one will inevitably sacrifice the other one. In this way, no matter which BS method is used, it is bound to increase the trouble of data processing. In addition, the BS must be used in the specific application, and the performance of the BS needs to be verified by the specific application. For example, target detection is performed by using the image which is synthesized by using the optimal segment subset obtained by a BS method. The effectiveness of this method in target detection can only be verified by analyzing the detection effect.

To attack these problems, this article first proposes sequential band fusion (SBF) for hyperspectral target detection which is different from the traditional BS methods. First, SBF is not necessary to determine the number of bands selected (n_{BS}), which can only be estimated by the additional algorithm in advance. Second, SBF does not need to design an algorithm to search for the optimal solution with high performance. The fusion process is dynamic. The real-time effect of the band on the application effect can be observed after each band fused. For example, SBF can be used in target detection to obtain the variation trend of detection values. The influence of each band on the detection can be obtained by analyzing the variation trend. Third, the fusion process of SBF is controllable. In BS, once the optimal solution is selected as a few bands, other bands will be discarded.

To avoid the problems existing in BS, there are two main data processing methods: progressive and recursive. Progressive band processing (PBP) is widely used in applications. For example, progressive BS was proposed to compress and transmit data progressively [1]. In addition, spectral unmixing can be performed progressively by selecting different bands for various endmembers [2]. Progressive dimensionality reduction by transformation was particularly designed to perform data dimensionality reduction in terms of progressive information preservation [3]. The progressive band dimensionality process can be used to perform progressive band dimensionality expansion and reduction [4]. In addition, PBP can be combined with different algorithms for specific applications. For instance, PBP is combined with constrained energy minimization (CEM), reed-xiaoli detector (RXD), and linear spectral unmixing to detect targets and anomalies and process data unmixing [5]–[7]. Fast iterative pixel purity index combined with PBP can be processed band by band, while band acquisition is ongoing [8]. It is worth noting that the PBP can process in real-time which cannot be realized in BS [9], [10]. There are also many applications of recursive band processing. Recursive band processing combined with orthogonal subspace projection can be implemented band by band recursively without waiting for data being completely collected [11]. Recursive band processing combined with the automatic target generation process is used to find endmembers [12]. These methods usually deal with data according to band sequence, which contributes to real-time data processing. However, the fixed processing sequence limits its dynamic and adjustable in specific applications.

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The basic condition for the implementation of band fusion is that the fusion result must be consistent with the result of directly applying the corresponding band. In this article, two kinds of band fusion are proposed for hyperspectral target detection, the fusion between the single band and the fusion between band set and a single band. To realize band fusion, the detailed derivation of fusion theory is given.

To dynamically observe the real-time influence of band fusion on application effect, four versions of SBF are proposed in this article. Every fusion version not only avoids the disadvantages of determining n_{BS} and searching the optimal solution in BS, but also can dynamically display the application results of each band after fusion, so as to ensure that users can check the influence of a certain band on a specific application in real-time.

The first fusion method is SBF based on band sequence (SBF-BSQ), which is similar to PBP-CEM [5]. Fuse band one by one starting from the first band $\mathbf{b}_1$ according to the band sequential data acquisition format used by a hyperspectral imaging sensor. The impact of each band on the application results can be observed in real-time. Until all the bands are fused to get the final result. As a matter of fact, PBP-CEM can be considered as a special case of SBF presented in Section II-B, which fuses one single band currently being acquired by BSQ with a band subset that comprises previously added bands. Several applications have been also investigated for using PBP in [1]–[8].

The second method is SBF driven by initial band (SBF-IBD), which considers that the application effect of the initial band is very important. Therefore, SBF-IBD does not directly use $\mathbf{b}_1$ as the initial band for application like SBF-BSQ, but uses the optimal band selected by band prioritization criteria as the initial band, and takes the initial band as the benchmark for the subsequent band fusion.

Not satisfied with improving the application effect only by optimizing the initial band, we propose the third band fusion method, i.e., SBF based on band priority (SBF-BP), which can be regarded as an expansion of SBF-IBD. First, the whole bands are sorted using two BP criteria, called minimal variance-based BP (MinV-BP) and maximal variance-based BP (MaxV-BP), and then the band is fused according to the sort result [13]. The principle of SBF-BP is to improve the application effect by optimizing the entire fusion order.

All the above three methods need to fuse all bands. Different from them, the fourth method, i.e., SBF based on band selection (SBF-BS) only needs to fuse partial band. First, an n_{BS} is prepared for SBF-BS as *a priori* knowledge, and then fuse bands selected for specific applications according to the n_{BS}. Because the application effect can be constantly updated with the band fusion. In theory, SBF-BS can increase the efficiency of application.

The following is the innovation of this article.

1) Band fusion does not need to design search strategy, which reduces the workload of data processing and improves efficiency.
2) In band fusion, the band fusion and the application verification are run in parallel. In BS, the band is selected first, and then verified by the application. These two processes are independent and separate, which leads to the delay of the verification. The worst of it is that the BS process has to start all over again if the application does not perform well.
3) Unlike the BS directly discard the unselected bands, band fusion can dynamically adjust the fusion process. In other words, the band to be fused is not limited by the band that has been fused, and the fusion of subsequent bands is adjustable.
4) Band fusion can provide the variation of application effect, and the interested band can be analyzed according to the variation trend.

This article is arranged as follows: to ensure the accuracy of detection, Section II introduces the theoretical derivation of band fusion, including the fusion between single bands and the fusion between band sets and single bands. Section III proposes four versions of band fusion, namely SBF-BSQ, SBF-IBD, SBF-BP, and SBF-BS. Sections IV and V list the experiments of target detection on two hyperspectral datasets via band fusion and decision fusion and analyze the experimental data and results. Finally, Section VII concludes the article.

II. Band Fusion

CEM is one of the commonly used target detection algorithms. To verify the application effect of band fusion in target detection, this section briefly reviews its concept as follows.

Suppose that a hyperspectral image is represented by a collection of image pixel vectors, denoted by $\{\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N\}$, where $\mathbf{r}_i = (r_{i1}, r_{i2}, \ldots, r_{iL})^T$ for $1 \leq i \leq N$ is an L-dimensional pixel vector, N is the total number of pixels in the image, and L is the total number of spectral bands. Furthermore, assume that $\mathbf{d} = (d_1, d_2, \ldots, d_L)^T$ is specified by a desired signature of interest to be used for target detection. It should be noted that $\mathbf{d}$ does not have to be a data sample vector and can be a spectral signature chosen from a database or spectral library. The goal is to find a target detector that can detect data samples specified by the desired target signature $\mathbf{d}$ via a finite impulse response (FIR) linear filter with L filter coefficients, $\{\omega_1, \omega_2, \ldots, \omega_L\}$, denoted by an L-dimensional vector $\mathbf{w} = (\omega_1, \omega_2, \ldots, \omega_L)^T$ which minimizes the filter output energy subject to the constraint $\mathbf{d}^T\mathbf{w} = \mathbf{w}^T\mathbf{d} = 1$. Let y_i denote the output of the designed FIR filter resulting from the input $\mathbf{r}_i$. Then, y_i can be expressed by

$$y_i = \sum_{l=1}^{L} \omega_l r_{il} = (\mathbf{w})^T \mathbf{r}_i = \mathbf{r}_i^T \mathbf{w} \tag{1}$$

and the average energy of the filter output is given by

$$\begin{aligned}(1/N)\sum_{i=1}^{N} y_i^2 &= (1/N)\sum_{i=1}^{N} \left(\mathbf{r}_i^T \mathbf{w}\right)^2 \\ &= \mathbf{w}^T \left[(1/N)\sum_{i=1}^{N} \mathbf{r}_i \mathbf{r}_i^T\right] \mathbf{w} = \mathbf{w}^T \mathbf{R} \mathbf{w}\end{aligned} \tag{2}$$

where $\mathbf{R} = (1/N)\sum_{i=1}^{N} \mathbf{r}_i \mathbf{r}_i^T$ is the sample autocorrelation matrix of the image. The goal is to solve the following linearly

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constrained optimization problem:

$$\min_{\mathbf{w}}\{\mathbf{w}^T\mathbf{R}\mathbf{w}\} \quad \text{s.t. } \mathbf{d}^T\mathbf{w}=\mathbf{w}^T\mathbf{d}=1 \tag{3}$$

where $\mathbf{w}^T\mathbf{R}\mathbf{w}$ can be considered as either the variance resulting from signals not passing through the filter. The optimal solution to (3) is shown in [14]–[17] to be

$$\mathbf{w}^{\mathrm{CEM}}=(\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d})^{-1}\mathbf{R}^{-1}\mathbf{d} \tag{4}$$

and

$$\min_{\omega}\mathbf{w}\mathbf{R}^{-1}\mathbf{w}=(\mathbf{w}^{\mathrm{CEM}})^T\mathbf{R}^{-1}\mathbf{w}^{\mathrm{CEM}}=(\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d})^{-1} \tag{5}$$

which is the minimal variance resulting from unwanted signal sources impinging upon an array of sensors [18], referred to as CEM error. With the optimal weight, $\mathbf{w}^{\mathrm{CEM}}$ specified by (4) a filter called CEM, denoted by $\delta^{\mathrm{CEM}}(\mathbf{r})$ was derived [14] as a target detector specified by

$$\delta^{\mathrm{CEM}}(\mathbf{r})=(\mathbf{w}^{\mathrm{CEM}})^T\mathbf{r}=(\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d})^{-1}(\mathbf{R}^{-1}\mathbf{d})^T\mathbf{r}. \tag{6}$$

A. Band Fusion of Two Single Bands

The fusion between single bands can be regarded as a special case of the fusion of band set and single band, that is, there is only one band in the band set. This special situation always occurs at the beginning of the fusion.

Let $\mathbf{r}_{\mathbf{b}_s}=(r_{1s},r_{2s},\ldots,r_{Ns})^T$ and $\mathbf{r}_{\mathbf{b}_t}=(r_{1t},r_{2t},\ldots,r_{Nt})^T$ be band vectors of two single band images $\mathbf{b}_s$ and $\mathbf{b}_t$ at spectral wavelengths s and t, respectively. Also, let assume that the data matrix formed by these two band vectors is given by

$$\mathbf{X}_{\mathbf{b}_s\mathbf{b}_t}=\begin{bmatrix} r_{1s} & r_{2s} & \cdots & r_{(N-1)s} & r_{Ns} \\ r_{1t} & r_{2t} & \cdots & r_{(N-1)t} & r_{Nt} \end{bmatrix} \tag{7}$$

$$\begin{aligned}\mathbf{R}_{\mathbf{b}_s\mathbf{b}_t}&=(1/N)\mathbf{X}_{\mathbf{b}_s\mathbf{b}_t}\mathbf{X}^T_{\mathbf{b}_s\mathbf{b}_t}=N^{-1}\begin{bmatrix}\mathbf{r}^T_{\mathbf{b}_s}\\ \mathbf{r}^T_{\mathbf{b}_t}\end{bmatrix}\begin{bmatrix}\mathbf{r}_{\mathbf{b}_s} & \mathbf{r}_{\mathbf{b}_t}\end{bmatrix}\\ &=N^{-1}\begin{bmatrix}\mathbf{r}^T_{\mathbf{b}_s}\mathbf{r}_{\mathbf{b}_s} & \mathbf{r}^T_{\mathbf{b}_s}\mathbf{r}_{\mathbf{b}_t}\\ \mathbf{r}^T_{\mathbf{b}_t}\mathbf{r}_{\mathbf{b}_s} & \mathbf{r}^T_{\mathbf{b}_t}\mathbf{r}_{\mathbf{b}_t}\end{bmatrix}\end{aligned} \tag{8}$$

where

$$\mathbf{R}^{-1}_{\mathbf{b}_s\mathbf{b}_t}=N\begin{bmatrix}\mathbf{r}^T_{\mathbf{b}_s}\mathbf{r}_{\mathbf{b}_s} & \mathbf{r}^T_{\mathbf{b}_s}\mathbf{r}_{\mathbf{b}_s}\\ \mathbf{r}^T_{\mathbf{b}_t}\mathbf{r}_{\mathbf{b}_s} & \mathbf{r}^T_{\mathbf{b}_t}\mathbf{r}_{\mathbf{b}_s}\end{bmatrix}^{-1}. \tag{9}$$

In this case, when $\mathbf{r}$ and $\mathbf{R}$ use only two bands, $\mathbf{b}_s$ and $\mathbf{b}_t$, CEM fused by $\mathbf{b}_s$ and $\mathbf{b}_t$ can be specified by $(\mathbf{d}^T_{\mathbf{b}_s\mathbf{b}_t}\mathbf{R}^{-1}_{\mathbf{b}_s\mathbf{b}_t}\mathbf{d}_{\mathbf{b}_s\mathbf{b}_t})^{-1}(\mathbf{R}^{-1}_{\mathbf{b}_s\mathbf{b}_t}\mathbf{d}_{\mathbf{b}_s\mathbf{b}_t})^T\mathbf{r}_{\mathbf{b}_s\mathbf{b}_t}$ in terms of $(d^T_{\mathbf{b}_s}R^{-1}_{\mathbf{b}_s}d_{\mathbf{b}_s})^{-1}(R^{-1}_{\mathbf{b}_s}d_{\mathbf{b}_s})^Tr_{\mathbf{b}_s}$ and $(d^T_{\mathbf{b}_t}R^{-1}_{\mathbf{b}_t}d_{\mathbf{b}_t})^{-1}(R^{-1}_{\mathbf{b}_t}d_{\mathbf{b}_t})^Tr_{\mathbf{b}_t}$. $\mathbf{d}_{\mathbf{b}_s\mathbf{b}_t}=(d_{\mathbf{b}_s},d_{\mathbf{b}_t})^T$, $\mathbf{r}_{\mathbf{b}_s\mathbf{b}_t}=(r_{\mathbf{b}_s},r_{\mathbf{b}_t})^T$, $R_{\mathbf{b}_s}=(1/N)\sum_{i=1}^N r^2_{is}$ and $R_{\mathbf{b}_t}=(1/N)\sum_{i=1}^N r^2_{it}$. Appendix A proves that

$$\begin{aligned}\delta^{\mathrm{CEM}}_{\mathbf{b}_s\mathbf{b}_t}&=k^{-1}_{\mathbf{b}_s\mathbf{b}_t}k_{\mathbf{b}_s}\delta^{\mathrm{CEM}}_{\mathbf{b}_s}+\frac{1}{N}k^{-1}_{\mathbf{b}_s\mathbf{b}_t}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\left(d_{\mathbf{b}_s}\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}-Nd_{\mathbf{b}_t}\right)\\ &\quad\times\left(\upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}r_{\mathbf{b}_s}-Nr_{\mathbf{b}_t}\right)\end{aligned} \tag{10}$$

$$k_{\mathbf{b}_s\mathbf{b}_t}=k_{\mathbf{b}_s}+\frac{1}{N}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\left(d_{\mathbf{b}_s}\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}-Nd_{\mathbf{b}_t}\right)^2 \tag{11}$$

where $\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}=\mathbf{R}^{-1}_{\mathbf{b}_s}\mathbf{x}_{\mathbf{b}_s}\mathbf{x}_{\mathbf{b}_t}$ and $\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}=(\mathbf{x}^T_{\mathbf{b}_t}[P^{\perp}_{\mathbf{X}^T_{\mathbf{b}_s}}]\mathbf{x}_{\mathbf{b}_t})^{-1}$.

B. Band Fusion of One Band Subset With Single Band

The fusion of one band subset with a single band is the core step of sequential fusion processing. Let $\mathbf{B}_{s,m}=\{\mathbf{b}_{s_1},\mathbf{b}_{s_2},\ldots,\mathbf{b}_{s_m}\}$ be band set and its associated data matrix is defined by

$$\mathbf{X}_{\mathbf{B}_{s,m}}=\begin{bmatrix} r_{1s_1} & r_{2s_1} & \cdots & r_{(N-1)s_1} & r_{Ns_1} \\ r_{1s_2} & r_{2s_2} & \cdots & r_{(N-1)s_2} & r_{Ns_2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ r_{1s_{m-1}} & r_{2s_{m-1}} & \cdots & r_{(N-1)s_{m-1}} & r_{Ns_{m-1}} \\ r_{1s_m} & r_{2s_m} & \cdots & r_{(N-1)s_m} & r_{Ns_m} \end{bmatrix} \tag{12}$$

$$\mathbf{X}_{\mathbf{B}_{s,m}\mathbf{b}_t}=\begin{bmatrix} r_{1s_1} & r_{2s_1} & \cdots & r_{(N-1)s_1} & r_{Ns_1} \\ r_{1s_2} & r_{2s_2} & \cdots & r_{(N-1)s_2} & r_{Ns_2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ r_{1s_m} & r_{2s_m} & \cdots & r_{(N-1)s_m} & r_{Ns_m} \\ r_{1t} & r_{2t} & \cdots & r_{(N-1)t} & r_{Nt} \end{bmatrix} \tag{13}$$

$$\mathbf{R}_{\mathbf{B}_{s,m}\mathbf{b}_t}=N^{-1}\mathbf{X}_{\mathbf{B}_{s,m}\mathbf{b}_t}\mathbf{X}^T_{\mathbf{B}_{s,m}\mathbf{b}_t}=N^{-1}\sum_{i=1}^{N}\mathbf{X}_{\mathbf{B}_{s,m}}\mathbf{X}^T_{\mathbf{B}_{s,m}}+\mathbf{b}_t\mathbf{b}_t^T \tag{14}$$

where $\mathbf{b}_t=(r_{1t},r_{2t},\ldots,r_{Nt})^T$.

$$\mathbf{R}^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}=N\left(\sum_{i=1}^{N}\mathbf{X}_{\mathbf{B}_{s,m}}\mathbf{X}^T_{\mathbf{B}_{s,m}}+\mathbf{b}_t\mathbf{b}_t^T\right)^{-1}. \tag{15}$$

CEM fused by $\mathbf{B}_{s,m}$ and $\mathbf{b}_t$ can be specified by $(\mathbf{d}^T_{\mathbf{B}_{s,m}\mathbf{b}_t}\mathbf{R}^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}\mathbf{d}_{\mathbf{B}_{s,m}\mathbf{b}_t})^{-1}(\mathbf{R}^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}\mathbf{d}_{\mathbf{B}_{s,m}\mathbf{b}_t})^T\mathbf{r}_{\mathbf{B}_{s,m}\mathbf{b}_t}$ in terms of $(\mathbf{d}^T_{\mathbf{B}_{s,m}}\mathbf{R}^{-1}_{\mathbf{B}_{s,m}}\mathbf{d}_{\mathbf{B}_{s,m}})^{-1}(\mathbf{R}^{-1}_{\mathbf{B}_{s,m}}\mathbf{d}_{\mathbf{B}_{s,m}})^T\mathbf{r}_{\mathbf{B}_{s,m}}$ and $(d_t^TR_t^{-1}d_t)^{-1}(R_t^{-1}d_t)^Tr_t$ where $\mathbf{d}_{\mathbf{B}_{s,m}\mathbf{b}_t}=(\mathbf{d}^T_{\mathbf{B}_{s,m}},d_t)^T$, $\mathbf{r}_{\mathbf{B}_{s,m}\mathbf{b}_t}=(\mathbf{r}^T_{\mathbf{B}_{s,m}},r_t)^T$ and $R_t=(1/N)\sum_{i=1}^N r^2_{it}$. Appendix B proves that

$$\begin{aligned}\delta^{\mathrm{CEM}}_{\mathbf{B}_{s,m}\mathbf{b}_t}&=k^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}k_{\mathbf{B}_{s,m}}\delta^{\mathrm{CEM}}_{\mathbf{B}_{s,m}}\\ &\quad+\frac{1}{N}k^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\left(d^T_{\mathbf{B}_{s,m}}\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}-Nd_{\mathbf{b}_t}\right)\\ &\quad\times\left(\upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\mathbf{r}_{\mathbf{B}_{s,m}}-Nr_{\mathbf{b}_t}\right)\end{aligned} \tag{16}$$

$$k_{\mathbf{B}_{s,m}\mathbf{b}_t}=k_{\mathbf{B}_{s,m}}+\frac{1}{N}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\left(d^T_{\mathbf{B}_{s,m}}\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}-Nd_{\mathbf{b}_t}\right)^2 \tag{17}$$

where $\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}=\mathbf{R}^{-1}_{\mathbf{B}_{s,m}}\mathbf{X}_{\mathbf{B}_{s,m}}\mathbf{x}_{\mathbf{b}_t}$, and $\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}=(\mathbf{x}^T_{\mathbf{b}_t}[P^{\perp}_{\mathbf{X}^T_{\mathbf{B}_{s,m}}}]\mathbf{x}_{\mathbf{b}_t})^{-1}$.

III. SBF Algorithm

To verify the application effect of SBF in the target detection of hyperspectral images, this article proposes four band fusion methods, namely SBF-BSQ, SBF-IBD, SBF-BP, and SBF-BS. The algorithm design and processing flow of each method are introduced one by one as follows.

A. Sequential BF-BSQ

The first fusion method SBF-BSQ is the simplest one, that is, perform SBF based on BSQ of data collection. Receive from the first band $\mathbf{b}_1$ and fuse in turn according to the BSQ. Once a band is received, it will be fused for target detection

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immediately until the last band is fused to get the target detection results of all bands. The algorithm is described as follows.

SBF-BSQ

1. Initial condition:
   The initial band to be collected is $\mathbf{b}_1$ according to BSQ, denoted by $\mathbf{b}_{\text{initial}}^{\text{BSQ}} = \mathbf{b}_1^{\text{BSQ}}$. Let $\mathbf{\Omega}$ denote the set of all L bands and let $\mathbf{\Omega}_k$ denote the set of k fused bands. $\mathbf{\Omega}_k^c = \mathbf{\Omega} - \mathbf{\Omega}_k$ denote the set of bands to be fused.
2. Fuses the 1st band $\mathbf{b}_1$ with the second band $\mathbf{b}_2$ by (10) to get the fusion result $\mathbf{R}_2$. $\mathbf{\Omega}_k = \{\mathbf{b}_1, \mathbf{b}_2\}$.
3. For the kth band $\mathbf{b}_k$ to be fused with $k \geq 3$, fuses $\mathbf{b}_k$ by (16) according to BSQ to get the fusion result $\mathbf{R}_k$. $\mathbf{\Omega}_k = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k\}$.
4. Check if $k = L$. If yes, the algorithm is terminated. Otherwise, increase k by one, $k \leftarrow k + 1$ and go to step 3.

Since the fusion is based on BSQ, SBF-BSQ can detect in real-time. Once a band is collected, it is immediately fused for target detection without waiting for all the bands to be received before starting detect. Data collection, band fusion, and target detection are run in parallel. Thus, the fusion target detection results are obtained when all data are collected. SBF-BSQ can effectively improve the detection efficiency while ensuring the accuracy of the detection effect.

B. SBF-IBD

Different from using $\mathbf{b}_1$ as the initial band directly in SBF-BSQ, the SBF-IBD uses a certain band priority criterion to determine which one can be selected as the initial band. As the first fused band, the detection ability of the initial band has a crucial impact on the whole subsequent target detection process [19]. According to (5), for each single band image $\mathbf{b}_l$, we calculate the variance by MinV-BP and MaxV-BP [13].

1) MinV-BP:

$$V(\mathbf{b}_l) = \left(\mathbf{d}_{\mathbf{b}_l}^T \mathbf{R}_{\mathbf{b}_l}^{-1} \mathbf{d}_{\mathbf{b}_l}\right)^{-1}. \tag{18}$$

The full band set Ω in (5) was replaced with the lth band $\{\mathbf{b}_l\}$ of the hyperspectral image in (18). For each band, a $V(\mathbf{b}_l)$ can be calculated. $V(\mathbf{b}_l)$ can be used to measure the priority of each band. The smaller $V(\mathbf{b}_l)$ is, the higher the priority of the band $\mathbf{b}_l$ is.

2) MaxV-BP:

$$V(\Omega - \mathbf{b}_l) = \left(\mathbf{d}_{\Omega-\{\mathbf{b}_l\}}^T \mathbf{R}_{\Omega-\{\mathbf{b}_l\}}^{-1} \mathbf{d}_{\Omega-\{\mathbf{b}_l\}}\right)^{-1}. \tag{19}$$

Analogous to MinV-BP, the full band set Ω in (5) was replaced with $\{\Omega - \mathbf{b}_l\}$ in (19). $V(\Omega - \mathbf{b}_l)$ denotes the variance of the full band set except the lth band $\{\mathbf{b}_l\}$. For each band, a $V(\Omega - \mathbf{b}_l)$ can be calculated. $V(\Omega - \mathbf{b}_l)$ can also be used to measure the priority of each band. The larger $V(\Omega - \mathbf{b}_l)$ is, the greater the variance of the band set which removed the $\mathbf{b}_l$ is, and the higher the priority of the band $\mathbf{b}_l$ is. The SBF-IBD is designed as follows.

SBF-IBD

1. Find the initial band by
$$\mathbf{b}_{initial} = \mathbf{b}_l^* = \arg\left\{\min_{\mathbf{b}_l \in \Omega} \mathrm{V}(\mathbf{b}_l)\right\} \tag{20}$$
or
$$\mathbf{b}_{initial} = \mathbf{b}_l^* = \arg\left\{\max_{\mathbf{b}_l \in \Omega} \mathrm{V}(\Omega - \mathbf{b}_l)\right\} \tag{21}$$
2. For the kth band $\mathbf{b}_k$ to be fused where $k \geq 2$, then fuses previous band by (10) or (16) in a backward and circular manner to run through all bands, $\mathbf{b}_k = \mathbf{b}_{l-1}$ and obtains fusion result $\mathbf{R}_k$.
3. Check if $\mathbf{b}_k = \mathbf{b}_1$. If yes, go to step 4. Otherwise, increase k by one, $k \leftarrow k + 1$ and decrease l by one, $l \leftarrow l - 1$ and go to step 2.
4. For the kth band to be fused where $k > l$, then fusion from the last band $\mathbf{b}_L$ in a backward and circular manner, $\mathbf{b}_k = \mathbf{b}_l = \mathbf{b}_L$ to obtain fusion result $\mathbf{R}_k$ until it reaches the next band of $\mathbf{b}_l^*$, $\mathbf{b}_{l+1}^*$.
5. Check if $\mathbf{b}_k = \mathbf{b}_{l+1}^*$. If yes, the algorithm is terminated. Otherwise, increase k by one, $k \leftarrow k + 1$ and decrease l by one, $l \leftarrow l - 1$ and go to step 4.

C. SBF-BP

To study the relationship between the order of band fusion and the result of target detection, SBF-BP first sorts all bands using BP criterion, and then fuses the bands in turn according to the sequencing result for target detection. Two types of BP, single band MinV-BP and leave-one-out MaxV-BP are used for SBF-BP [13]. The SBF-BP can be defined as follows.

SBF-BP

1. Rank all bands for $1 \leq l \leq L$ according to $\{\mathrm{V}(\mathbf{b}_l)\}_{l=1}^L$ by (18) or $\{\mathrm{V}(\Omega - \mathbf{b}_l)\}_{l=1}^L$ by (19)
$$\mathbf{b}_j \succ \mathbf{b}_k \Leftrightarrow \mathrm{V}(\mathbf{b}_j) < \mathrm{V}(\mathbf{b}_k) \tag{22}$$
or
$$\mathbf{b}_j \succ \mathbf{b}_k \Leftrightarrow \mathrm{V}(\Omega - \mathbf{b}_j) > \mathrm{V}(\Omega - \mathbf{b}_k) \tag{23}$$
where the notation "$\succ$" is used to indicate "superior to." The final band sequence is $\{\mathbf{b}_{l_1}, \mathbf{b}_{l_2}, \ldots, \mathbf{b}_{l_L}\}$.
2. Fuses $\mathbf{b}_{l_1}$ with $\mathbf{b}_{l_2}$ by (10) to get the fusion result $\mathbf{R}_2$.
3. For the kth band $\mathbf{b}_k$ to be fused where $k \geq 3$, then fuses the $\mathbf{b}_{l_k}$ by (16) according to the band sequence ordered by two BP criterion separately until it reaches the last band $\mathbf{b}_{l_L}$.
4. Check if $k = L$. If yes, the algorithm is terminated. Otherwise, increase k by one, $k \leftarrow k + 1$ and go to step 3.

D. SBF-BS

To study the influence of band and its number on target detection results, SBF-BS uses BS algorithm to select a certain

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(a) (b)

Fig. 1. (a) Airport scene which contains 13 targets. (b) Ground truth map of pixel spatial locations of 13 interested targets.

number of bands for fusion detection experiment. Two types of BS, single band SF-CTBS and SB-CTBS are used for SBF-BS [13].

SBF-BS

1. Initial condition:
   Let n_{BS} be the number of bands needed to be selected, which can be determined by a criterion such as virtual dimensionality (VD).
2. Band selection:
   Select band according to n_{BS} by

$$\mathbf{b}_{l_k} = \arg\left\{ \min_{\mathbf{b}_l \in \Omega^c_{k-1}} \left(\mathbf{d}^T_{\Omega_{k-1}\cup\{\mathbf{b}_l\}} \mathbf{R}^{-1}_{\Omega_{k-1}\cup\{\mathbf{b}_l\}} \mathbf{d}_{\Omega_{k-1}\cup\{\mathbf{b}_l\}} \right)^{-1} \right\} \tag{24}$$

   or

$$\mathbf{b}_{l_k} = \arg\left\{ \max_{\mathbf{b}_l \in \Omega^c_{k-1}} \left(\mathbf{d}^T_{\Omega-(\Omega_{k-1}\cup\{\mathbf{b}_l\})} \mathbf{R}^{-1}_{\Omega-(\Omega_{k-1}\cup\{\mathbf{b}_l\})} \times \mathbf{d}^T_{\Omega-(\Omega_{k-1}\cup\{\mathbf{b}_l\})} \mathbf{R}^{-1}_{\Omega-(\Omega_{k-1}\cup\{\mathbf{b}_l\})} \right)^{-1} \right\} \tag{25}$$

3. Band fusion:
   Fuse the kth band $\mathbf{b}_{l_k}$ to be fused where $k \geq 2$ by (10) or (16) and obtains fusion result $\mathbf{R}_k$.
4. Check if $k = n_{\mathrm{BS}}$. If yes, the algorithm is terminated. Otherwise, increase k by one, $k \leftarrow k + 1$ and go to step 3.

IV. Discussions on Role of SBF

Intuitively, it is possible to consider SBF as an extension of the BS method, but this is wrong. SBF and BS are two different band processing methods and have great differences in target design, data processing, and algorithm structure.

A. Purpose of SBF

There are two fundamental problems in the BS method. First, n_{BS} needs to be known as prior knowledge before selecting bands. However, the calculation of n_{BS} usually requires the spectral information of hyperspectral images, which will undoubtedly increase the storage burden of data processing system [10, Chapter 4]. Second, how to design an optimal solution search algorithm with convergence. Although the

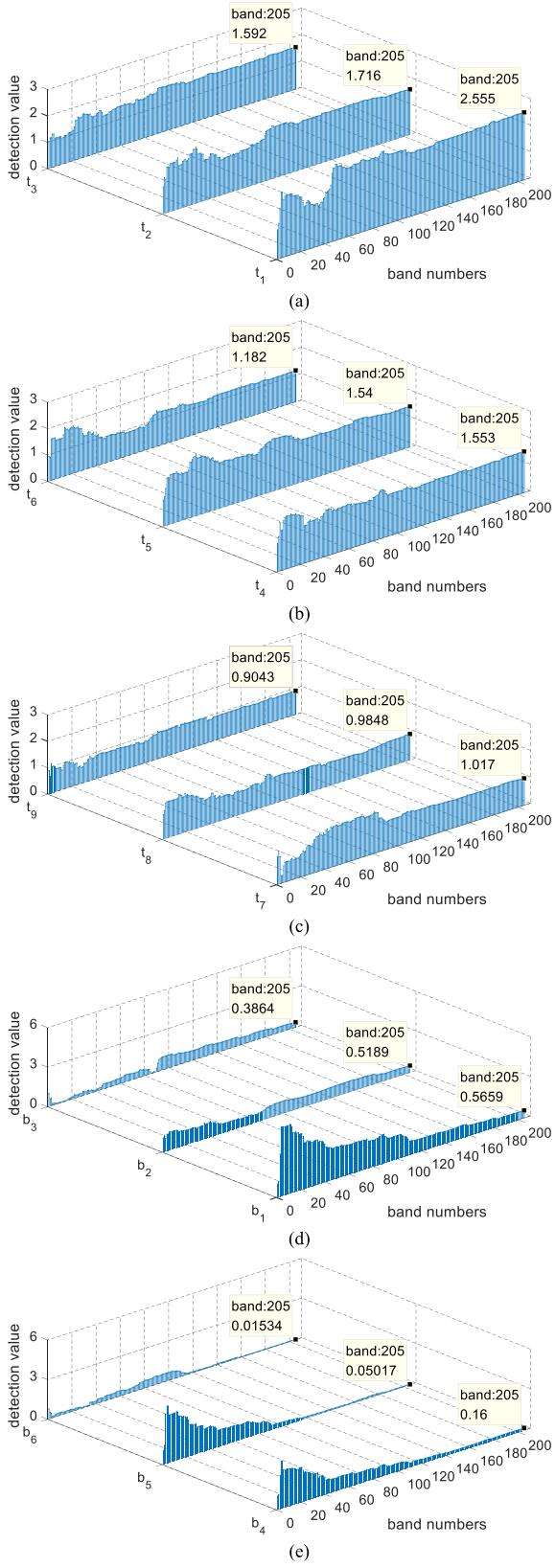

Fig. 2. Variation of detection values of nine target pixels and six background pixels detected by CEM based on SBF-BSQ. (a)–(c) Variation of nine target pixels. (d), (e) Variation of six background pixels.

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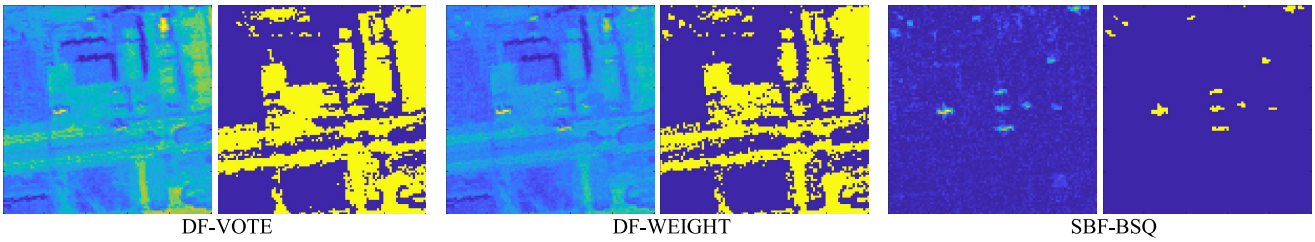

Fig. 3. Detection maps by CEM based on DF-VOTE, DF-WEIGHT, and SBF-BSQ as along with binary map obtained by Otsu's method.

TABLE I
AUC VALUES OF DETECTION RESULTS BY DF-VOTE, DF-WEIGHT, AND SBF-BSQ

n_{BS}	Band subsets	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{ODP}	AUC_{SNPR}
14	DF-VOTE	0.7473	**0.3088**	0.1956	1.0561	0.5517	0.1132	0.8605	1.5789
	DF-WEIGHT	0.8496	0.2558	0.1226	1.1055	0.7271	0.1333	0.9829	2.0875
	SBF-BSQ	**0.9015**	0.2256	**0.0611**	**1.1271**	**0.8404**	**0.1645**	**1.0660**	**3.6925**
28	DF-VOTE	0.7676	**0.3746**	0.2375	1.1422	0.5300	0.1371	0.9046	1.5770
	DF-WEIGHT	0.8532	0.3109	0.1526	**1.1641**	0.7006	0.1584	1.0115	2.0380
	SBF-BSQ	**0.8655**	0.2789	**0.0687**	1.1444	**0.7967**	**0.2102**	**1.0757**	**4.0586**
42	DF-VOTE	0.8069	**0.4064**	0.2635	1.2133	0.5434	0.1429	0.9498	1.5421
	DF-WEIGHT	0.8945	0.3438	0.1720	1.2383	0.7225	0.1719	1.0663	1.9994
	SBF-BSQ	**0.9221**	0.3332	**0.0712**	**1.2553**	**0.8510**	**0.2620**	**1.1841**	**4.6815**
205	DF-VOTE	0.5250	0.3997	0.3701	0.9246	0.1549	0.0296	0.5545	1.0799
	DF-WEIGHT	0.7582	**0.4463**	0.2892	1.2044	0.4690	0.1570	0.9152	1.5430
	SBF-BSQ	**0.9555**	0.3374	**0.0295**	**1.2929**	**0.9260**	**0.3079**	**1.2634**	**11.4373**

BS method is intended to solve this problem, it cannot traverse all possible band combinations. The band set obtained by the BS method is not an optimal result selected from all possible combinations, but a better combination selected from a limited number of possible combinations. In addition, the effect of BS needs to be verified by specific applications. BS and application verification cannot be carried out synchronously, which will undoubtedly reduce the efficiency of data processing. These shortcomings limit the application of the BS method, especially in some demanding data processing systems. For example, in the satellite data processing system, the amount of data to be processed is very large, but the data transmission and storage space of the satellite system are limited. Efficiently processing massive data to accurately detect targets is the foremost task of satellite target detection systems. To transmit information timely, accurately, and effectively, it is necessary to design a concise and efficient band processing algorithm, which is the motivation to propose SBF.

B. Difference of Four SBF Algorithms

To make SBF suitable for different application scenarios, four band fusion algorithms SBF-BSQ, SBF-IBD, SBF-BP, and SBF-BS are designed in this article. In the following, the application of SBF in the satellite target detection system is taken as an example to introduce these four fusion algorithms. SBF-BSQ is the simplest one, which directly fuses bands according to the acquisition sequence of the satellite system. After collecting each band, it is transmitted to the fusion mechanism for target detection. In the fusion mechanism, band fusion and target detection are run in parallel. Therefore, in SBF-BSQ, information acquisition, band fusion, and target detection operate synchronously, so as to achieve the real-time target detection of the satellite system. Different from real-time performance of SBF-BSQ, SBF-IBD, and SBF-BP are developed to perform band processing with BP. In these two algorithms, the bands need to be sorted before being input into the fusion mechanism. This makes these two methods cannot achieve real-time detection, but their advantage is that they can design different BP criteria for different application purposes, and improve the application effect by changing the fusion order. In addition, although SBF-IBD and SBF-BP give up real-time, band fusion and target detection still operate synchronously, which will improve the data processing efficiency of the satellite system. Unlike SBF-IBD and SBF-BP, which need to sort and fuse all bands, the detection process of SBF-BS can be adjusted dynamically. In SBF-BS, BS, band fusion, and target detection run in parallel. Gradually, select a band and input it to the fusion mechanism for target detection. When the detection result becomes stable, the selection of the band is stopped and the whole detection process is ended. These four fusion algorithms can provide the changing trend of the target detection value in the whole band fusion process, through which we can get the influence of each band on the target detection effect, and end the detection process according to the convergence trend of the detection value, without the need to determine n_{BS}.

V. REAL IMAGE EXPERIMENTS

An airport scene shown in Fig. 1 is used to corroborate the theory of band fusion. The Airport scene was captured by the Airborne Visible/Infrared Imaging Spectrometer sensor in the range of 0.4–2.5 μm in Los Angeles [20]. After removing noisy bands, the data have 205 spectral bands and 100 $\times$ 100 pixels with a spatial resolution of 7.1 m.

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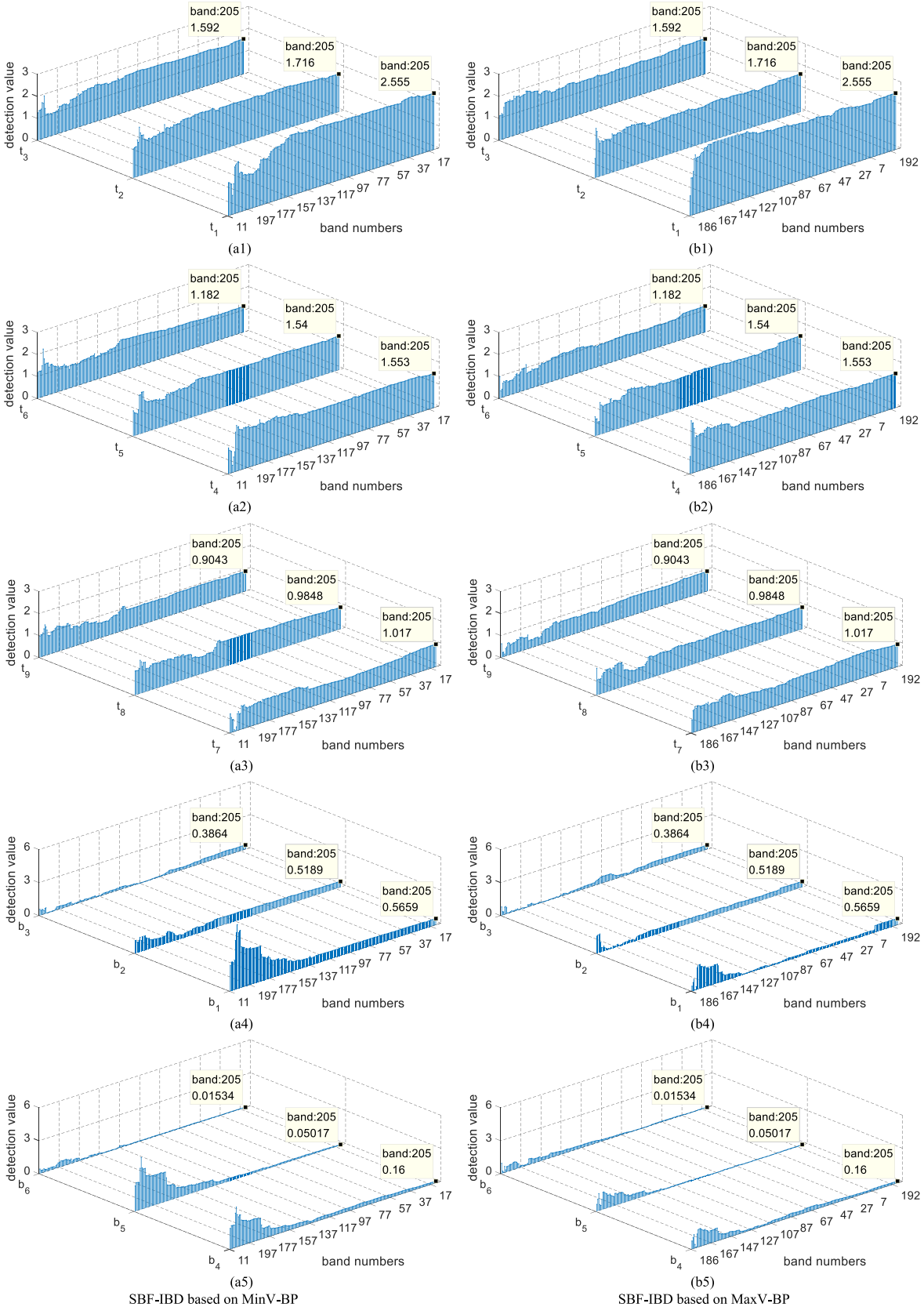

Fig. 4. Variations of detection value of nine target pixels and six background pixels based on SBF-IBD. (a1)–(a5) SBF-IBD based on MinV-BP. (b1)–(b5) SBF-IBD based on MaxV-BP.

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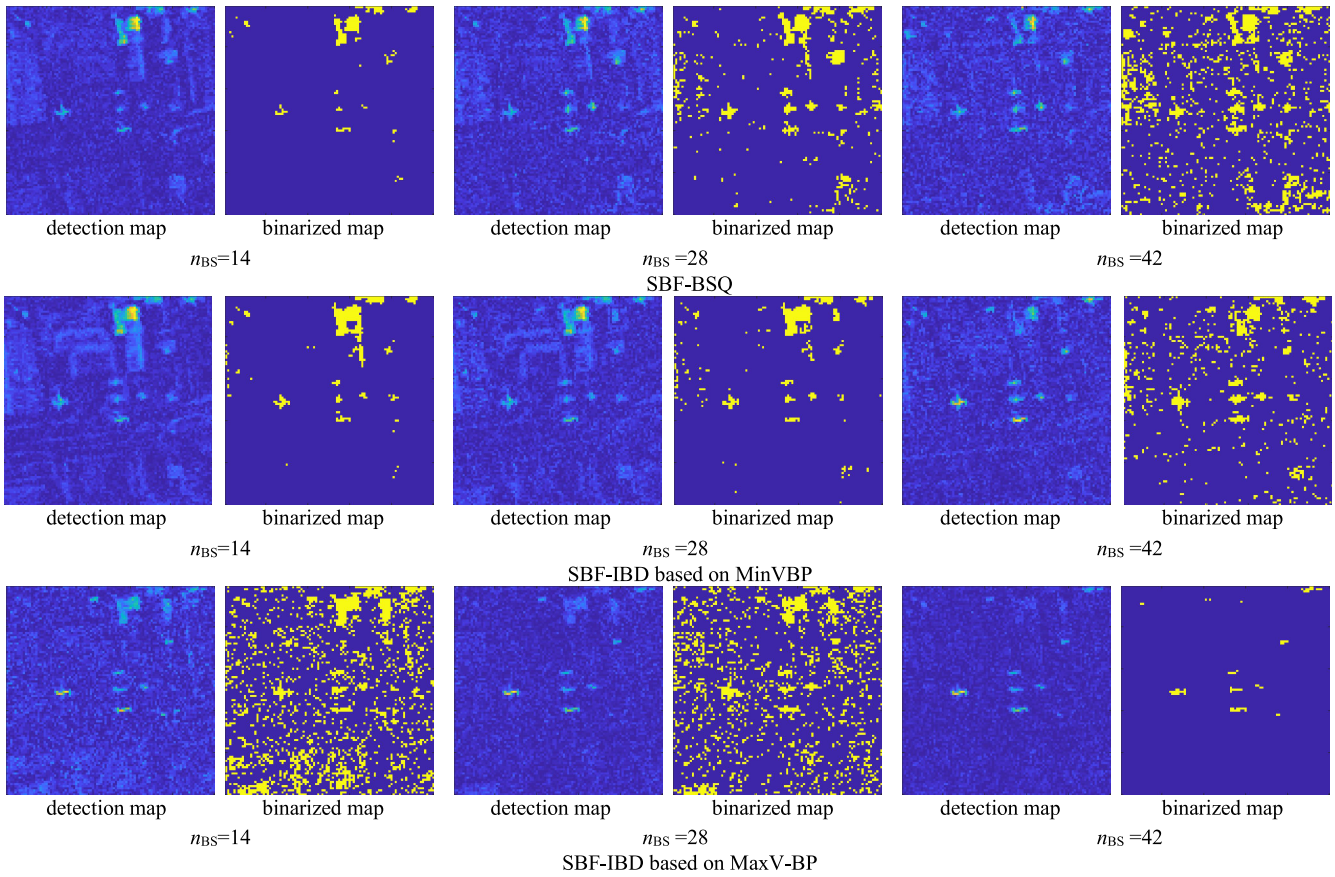

Fig. 5. Detection results and binarized image of SBF-BSQ, SBF-IBD based on MinVBP and MaxV-BP when $n_{BS} = 14$, 28, and 42.

Fig. 1(b) shows the ground truth image of the airport scene with 13 interested targets [21], [22].

In this section, the hyperspectral image target detection method was performed to evaluate the proposed SBF. To better analyze the experiment, the results of band fusion are compared with that of decision fusion. Decision fusion is the fusion of the output of each processor. In general, according to different fusion strategies, decision fusion algorithms can be divided into three categories: decision fusion based on voting strategy (DF-VOTE), decision fusion based on weighted strategy (DF-WEIGHT), and decision fusion based on statistical methods [23]–[26]. Duan *et al.* [23] combined classification with a majority voting fusion mechanism. The label of each pixel in each classification result map was considered as a vote, and the final label was determined by the maximum value of the vote. Soltani *et al.* [25] applied adaptive weighting to decision fusion. This method assigns greater weight for the data with high credibility, so as to ensure the credibility of the fusion data. The band fusion methods proposed in this article is compared with DF-VOTE and DF-WEIGHT. All experiments were run on a computer with Intel Core i7-6700, NVIDIA GeForce GTX 1050 Ti, 8 GB memory. the OS is Windows 10, the platform is MATLAB R2018a. The following experiments were designed and conducted:

1) performing SBF-BSQ starting with the 1st band;
2) performing SBF-IBD in a backward and circular fashion where the initial band were selected by (20) and (21);
3) performing SBF-BP using band sorted by BP criterion (22) and (23);
4) performing SBF-BS using band selected by (24) and (25).

A. Sequential BF According to BSQ

The first experiment performed SBF according to BSQ. The aircraft is regarded as an interested target, and the CEM detector is used to detect the target in the airport scene. When the first band is received, it is immediately used for target detection, that is, the reception of the second band runs in parallel with the target detection of the first band. When the second band is received, it is immediately fused with the existing detection results of the first band by (10) to obtain the fusion result. By analogy, the subsequent bands are received in turn and fused with the existing detection results by (16) to obtain real-time detection results, until all the bands are fused to obtain the final fusion result. It should be noted that the transmission of data, the detection of target, and the fusion of band are run in parallel. As shown in Fig. 2, the variation trend of the detection value of any pixel in the image during the fusion process can be observed. The influence of each fusion band on the detection value can be observed in real-time, and the corresponding band can be evaluated according to whether the influence is beneficial.

To study the effect of band-by-band fusion on target detection and background suppression, Fig. 2 shows the variation

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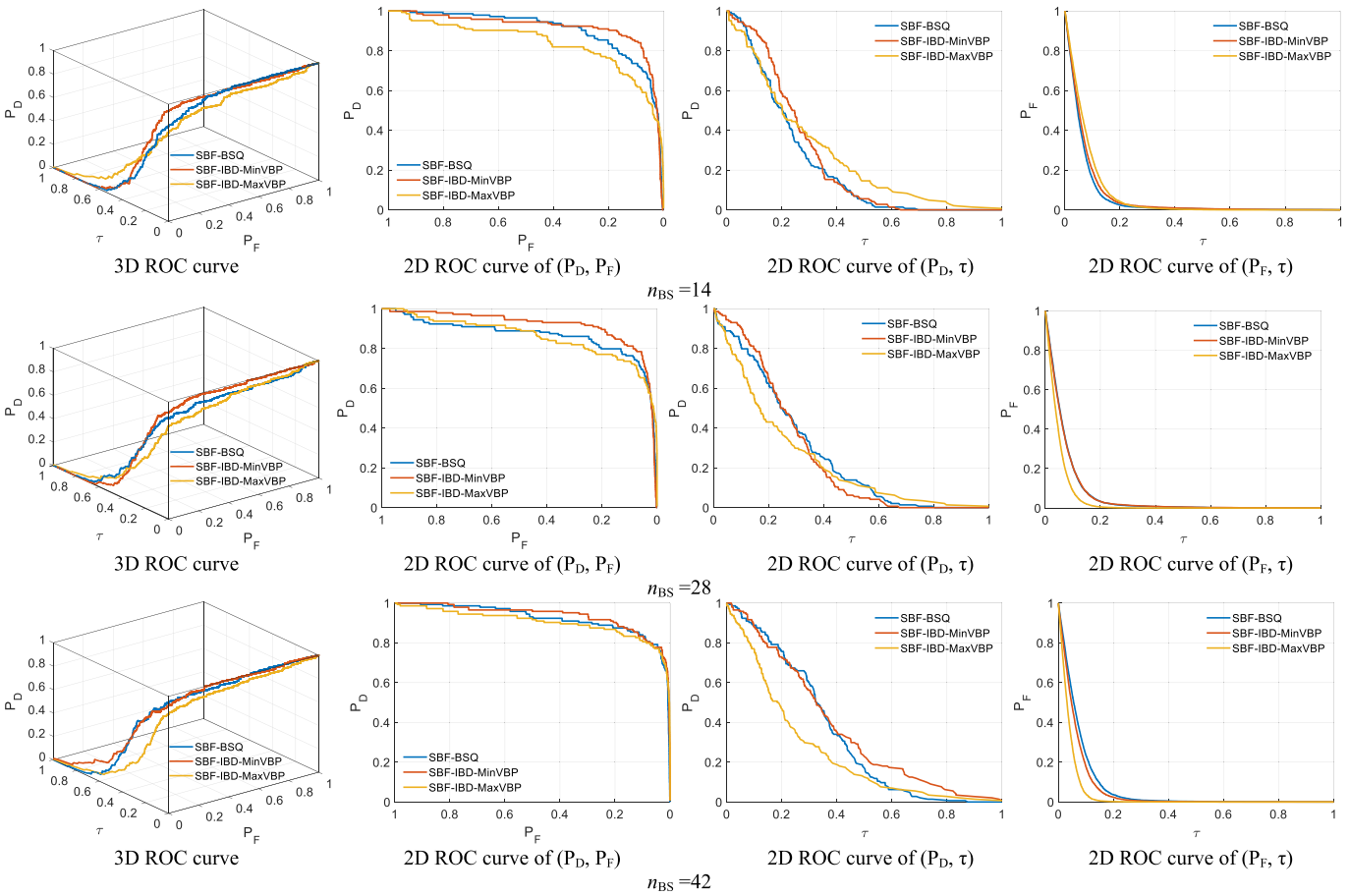

Fig. 6. 3-D ROC curves along with their generated three 2-D ROC curves of CEM detected target maps of SBF-BSQ, SBF-IBD-MinVBP, and SBF-IBD-MaxVBP.

TABLE II

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES OF DETECTION RESULTS USING DIFFERENT BAND SUBSETS

n_{BS}	Band subsets	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{ODP}	AUC_{SNPR}
14	DF-VOTE	0.3906	0.3127	0.3738	0.7033	0.0168	-0.0611	0.3295	0.8366
	DF-WEIGHT	0.6413	**0.4244**	0.3477	1.0656	0.2936	0.0767	0.7179	1.2205
	SBF-BSQ	0.9015	0.2256	**0.0611**	1.1271	0.8404	0.1645	1.0660	3.6925
	SBF-IBD-MinVBP	**0.9131**	0.2539	0.0679	**1.1671**	**0.8452**	0.1860	**1.0991**	3.7385
	SBF-IBD-MaxVBP	0.8352	0.2767	0.0735	1.1120	0.7617	**0.2032**	1.0385	**3.7654**
28	DF-VOTE	0.4040	0.3473	0.4047	0.7513	-0.0007	-0.0574	0.3466	0.8582
	DF-WEIGHT	0.6445	**0.4326**	0.3559	1.0771	0.2886	0.0768	0.7212	1.2157
	SBF-BSQ	0.8655	0.2789	0.0687	1.1444	0.7967	**0.2102**	1.0757	4.0586
	SBF-IBD-MinVBP	**0.9153**	0.2706	**0.0684**	**1.1859**	**0.8469**	0.2022	**1.1175**	3.9563
	SBF-IBD-MaxVBP	0.8600	0.2412	0.0460	1.1012	0.8140	0.1952	1.0552	**5.2438**
42	DF-VOTE	0.5353	0.4252	0.3855	0.9605	0.1498	0.0397	0.5750	1.1031
	DF-WEIGHT	0.7320	**0.4518**	0.2937	1.1838	0.4383	0.1581	0.8901	1.5385
	SBF-BSQ	0.9221	0.3332	0.0712	1.2553	0.8510	0.2620	1.1841	4.6815
	SBF-IBD-MinVBP	**0.9346**	0.3648	0.0609	**1.2994**	**0.8737**	**0.3039**	**1.2385**	5.9893
	SBF-IBD-MaxVBP	0.9023	0.2442	**0.0381**	1.1465	0.8643	0.2061	1.1085	**6.4151**
205	DF-VOTE	0.8111	0.3638	0.1934	1.1748	0.6177	0.1704	0.9814	1.8808
	DF-WEIGHT	0.9213	**0.3928**	0.1348	**1.3141**	0.7865	0.2580	1.1793	2.9134
	SBF-BSQ	**0.9555**	0.3374	**0.0295**	1.2929	**0.9260**	**0.3079**	**1.2634**	**11.4373**
	SBF-IBD-MinVBP	**0.9555**	0.3374	**0.0295**	1.2929	**0.9260**	**0.3079**	**1.2634**	**11.4373**
	SBF-IBD-MaxVBP	**0.9555**	0.3374	**0.0295**	1.2929	**0.9260**	**0.3079**	**1.2634**	**11.4373**

of detection values of nine target pixels and six background pixels, where Fig. 2(a)–(c) shows the variation of the target pixel, Fig. 2(d) and (e) shows that of the background pixel. As can be seen, neither the target pixel nor the background pixel, the variation of detection value is not monotonous. Moreover, the variation of the detection value is larger in the early stage of band fusion and smaller in the later stage. For example, the detection value of background pixels generally

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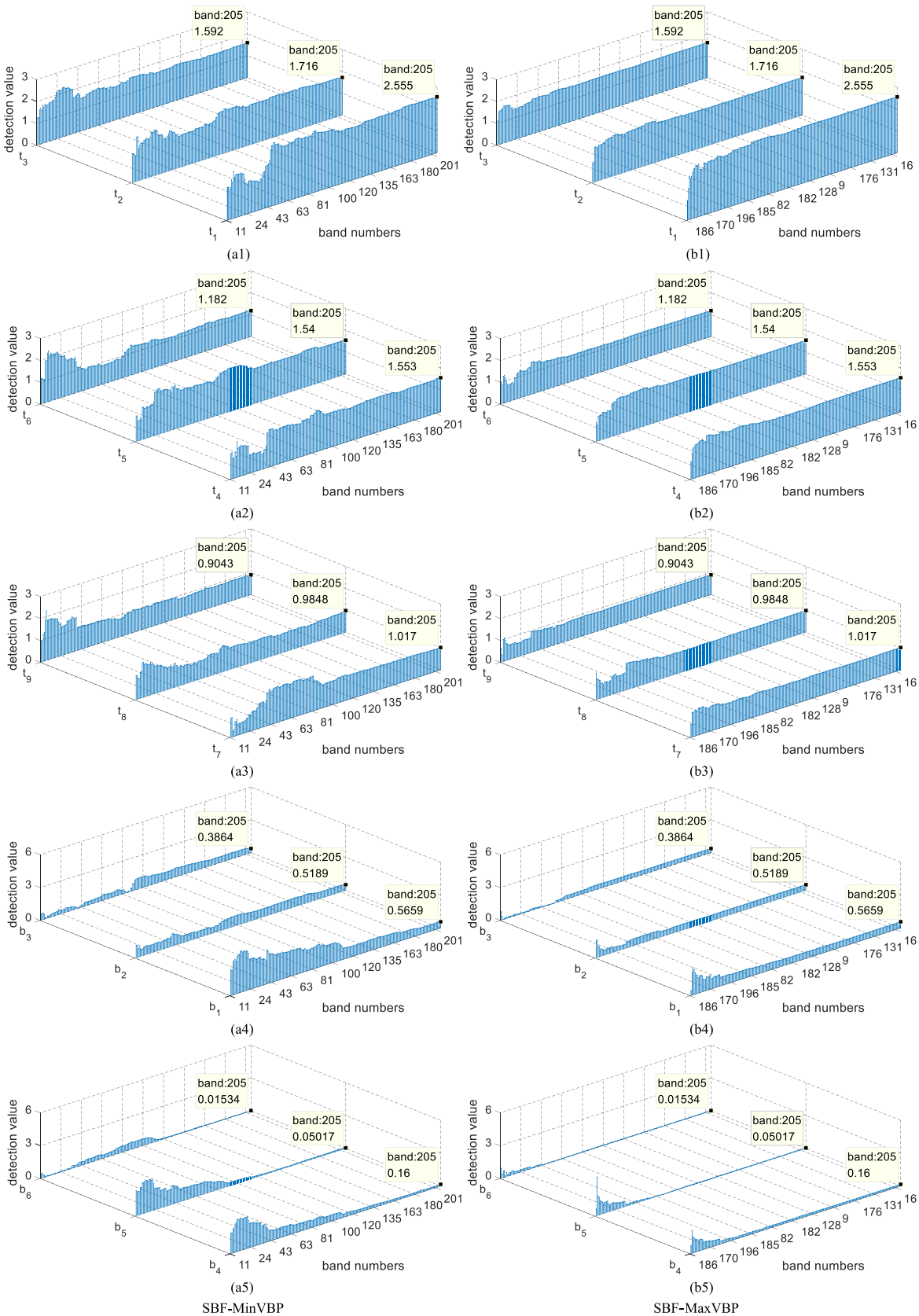

Fig. 7. Detected value fractions of nine target pixels and six background pixels produced by SBF-MinVBP and SBF-MaxVBP. (a1)–(a5) SBF-MinVBP. (b1)–(b5) SBF-MaxVBP.

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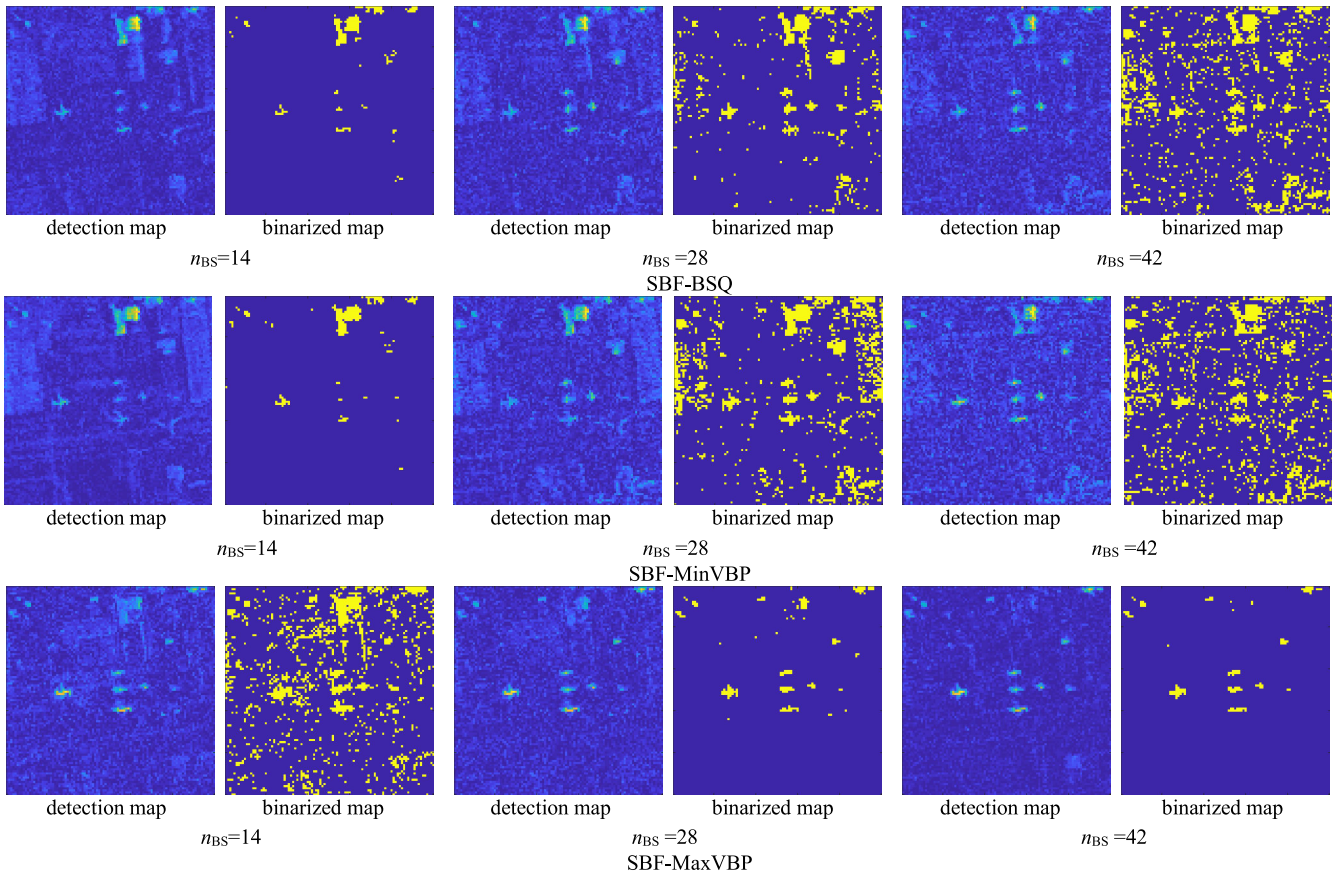

Fig. 8. Detection results and binarized image of SBF-BSQ, SBF-MinVBP, and SBF-MaxVBP when $n_{BS} = 14$, 28, and 42.

shows a downward trend, with a larger decline in the early stage of fusion and smaller in the later stage. This indicates that the detection tends to be stable in the later stage of band fusion.

It is worth noting that the detection results obtained by fusing all bands are consistent with the results of direct detection using all bands. Therefore, only the detection map and binary map of CEM using SBF-BSQ are shown in Fig. 3, which also verifies the SBF theory. The threshold for the binary map was calculated by the Otsu algorithm. The Otsu algorithm is an efficient algorithm for binarizing images based on the maximum variance between classes [27]. The n_{BS} of Airport scene was calculated as 14 by VD algorithm [10, Chapter 4], [28]–[31]. Table I listed the AUC values of detection results by DF-VOTE, DF-WEIGHT, and SBF-BSQ [32]. In addition to $AUC_{(PD,PF)}$, $AUC_{(PD,\tau)}$, and $AUC_{(PF,\tau)}$, several quantitative detection measures are further used to evaluate a detector's effectiveness, target detectability (TD), background suppressibility (BS), specifically, target detection in background (TDBS), overall detection probability (ODP), and signal-to-noise probability ratio (SNPR), all of which can effectively evaluate hyperspectral target detection quantitatively. The data in bold are the best AUC produced by DF-VOTE, DF-WEIGHT, and SBF-BSQ. As can be seen, the $AUC_{(D,F)}$, $AUC_{(F,\tau)}$, $AUC_{(BS)}$, $AUC_{(TDBS)}$, $AUC_{(ODP)}$, and $AUC_{(SNPR)}$ of SBF-BSQ are higher than DF-VOTE and DF-WEIGHT when $n_{BS} = 14$, 28, 42, and 205. This indicates that the fusion mechanism of SBF-BSQ is better and the fusion result is more accurate than DF-VOTE and DF-WEIGHT.

The advantages of SBF-BSQ are summarized as follows. First, SBF-BSQ can detect in real-time since the bands are fused according to BSQ. Second, data transmission, band fusion, and target detection are running in parallel, which improves detection efficiency. Third, the variation of the detection value of any pixel can be observed during the fusion process, so the detection ability of the band can be evaluated. Fourth, the fusion results are consistent with the results of direct detection, which ensures the accuracy of detection. In addition, band fusion does not need to control the number of bands used for detection by calculating n_{BS} in advance, but ends detection according to the variation trend of detection value.

B. SBF-IBD

Different from SBF-BSQ, SBF-IBD adjusts the fusion order by determining the initial band actively. Initial band is no longer the first band, but is replaced by band selected by BP criteria. The subsequent bands are fused based on this band. In SBF-IBD, the initial band is considered to be very important, and its change can directly affect the fusion process. In this experiment, the highest priority bands are band 11 and band 186 selected by (20) and (21). Then, band 11 and band 186 were used as the initial bands to fuse all bands

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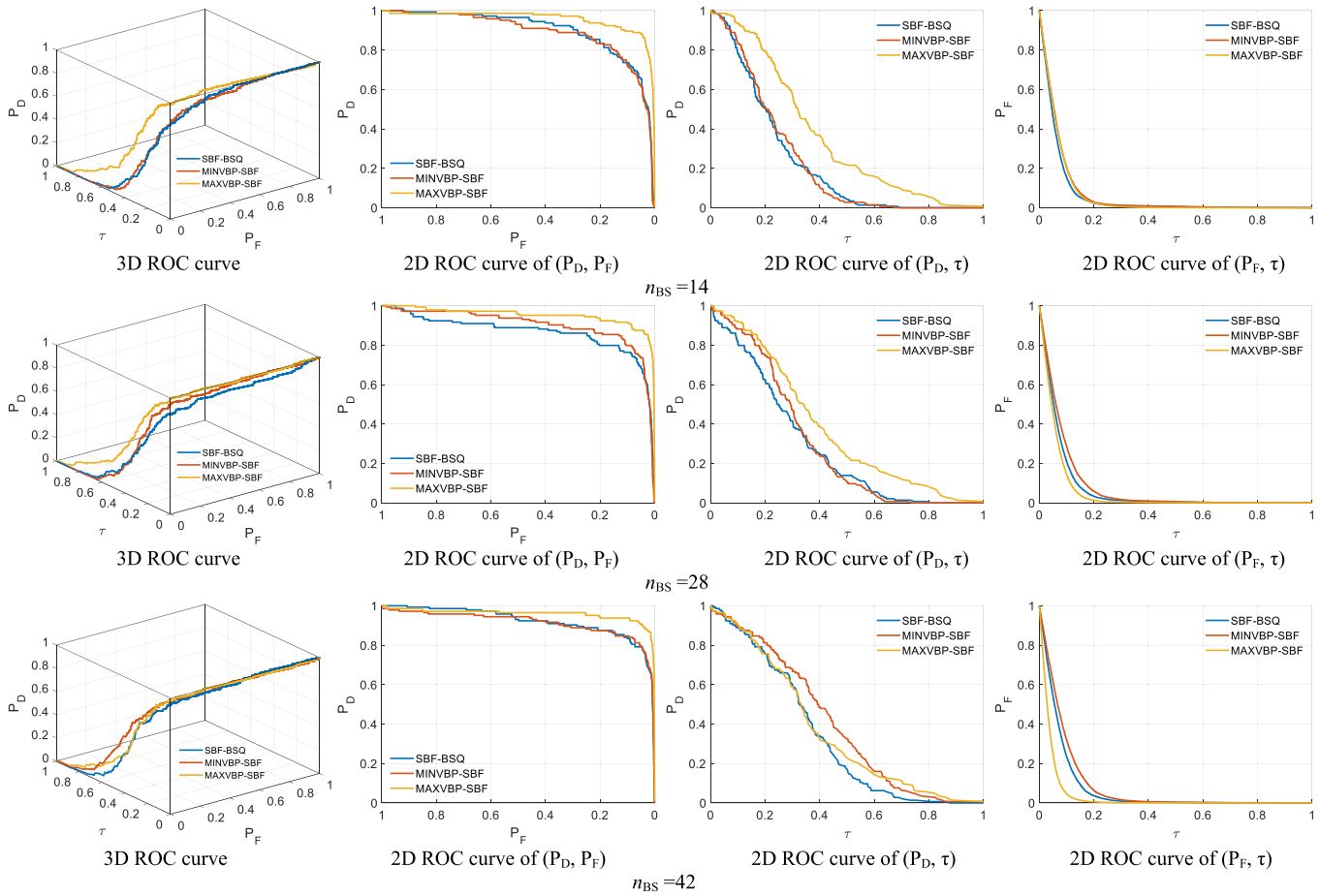

Fig. 9. 3-D ROC curves along with their generated three 2-D ROC curves of CEM detected target maps of SBF-BSQ, SBF-MinVBP, and SBF-MaxVBP.

in a backward manner to get the variation trend of detected values, as shown in Fig. 4.

In this experiment, the change of the initial band optimizes the initial conditions of the fusion. However, since the selection of the initial band needs to be combined with the band information of all bands, band fusion, and target detection can only be started after all bands have been received. This means that SBF-IBD cannot be detected in real-time like SBF-BSQ.

As can be seen from Fig. 4, like SBF-BSQ, the change of the detection value of SBF-IBD is also nonmonotonic. The changes in the early stage of the fusion are relatively large, and small in the later stage. Different from SBF-BSQ, SBF-IBD tends to be stable quickly. Especially when suppressing background pixels, the interference can be effectively suppressed by using a small number of bands. As shown in (d4), (e4), (d5), and (e5) of Fig. 4, the background pixels have been well suppressed and tend to be stable after only fusing 30 bands in SBF-IBD. However, the detection begins to stabilize only when nearly 100 bands are fused in SBF-BSQ.

The n_{BS} of Airport scene is calculated as 14 by VD algorithm [10, Chapter 4], [28]–[31]. To intuitively understand the impact of different fusion sequences on target detection, Fig. 5 shows the detection results of different methods when fusing the same number of bands. As can be seen, SBF-BSQ and SBF-IBD based on MinVBP can effectively suppress the interference of background on target detection when $n_{\mathrm{BS}} = 14$, while SBF-IBD based on MaxV-BP is not good enough. With the increasing number of fused bands, the background suppression effect of SBF-BSQ and SBF-IBD based on MinV-BP is decreasing, while suppression effect of SBF-IBD based on MaxV-BP is increasing.

The results in Fig. 5 only provided preliminary quantitative analysis. To conduct detailed quantitative studies a 3-D ROC analysis recently developed in [33]–[35] along with the three 3-D ROC curve-derived detection measures were used to perform detection analysis among the results in Fig. 5. Fig. 6 plotted their 3-D ROC curves along with their corresponding three 2-D ROC curves of Fig. 5 to evaluate their detection performances quantitatively. Table II lists the AUC values calculated from the three 2-D ROC curves of detection results using different band subsets. The results show that most AUC value of SBF-IBD-MinVBP were higher than DF-VOTE, DF-WEIGHT, SBF-BSQ, and SBF-IBD-MinVBP when $n_{\mathrm{BS}} =$ 14, 28, and 42. In particular, $\mathrm{AUC}_{(\mathrm{D,F})}$ of SBF-IBD-MinVBP has always been the highest. This indicates that SBF-IBD-MinVBP can not only accurately detect the target, but also effectively press the background. However, the above five methods are still not as good as using whole 205 bands for target detection.

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TABLE III

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES OF DETECTION RESULTS USING DIFFERENT BAND SUBSETS

n_{BS}	Band subsets	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{ODP}	AUC_{SNPR}
14	DF-VOTE	0.8112	0.3085	0.1773	1.1197	0.6339	0.1312	0.9424	1.7400
	DF-WEIGHT	0.8821	0.2941	0.1343	1.1762	0.7478	0.1598	1.0419	2.1904
	SBF-BSQ	0.9015	0.2256	**0.0611**	1.1271	0.8404	0.1645	1.0660	3.6925
	SBF-MinVBP	0.8901	0.2306	0.0690	1.1207	0.8212	0.1616	1.0517	3.3436
	SBF-MaxVBP	**0.9582**	**0.3668**	0.0648	**1.3250**	**0.8934**	**0.3021**	**1.2602**	**5.6647**
28	DF-VOTE	0.7889	**0.3826**	0.2396	1.1715	0.5493	0.1430	0.9319	1.5969
	DF-WEIGHT	0.8754	0.3535	0.1763	1.2289	0.6991	0.1773	1.0526	2.0057
	SBF-BSQ	0.8655	0.2789	0.0687	1.1444	0.7967	0.2102	1.0757	4.0586
	SBF-MinVBP	0.9007	0.2979	0.0806	1.1987	0.8202	0.2174	1.1181	3.6977
	SBF-MaxVBP	**0.9518**	0.3802	**0.0560**	**1.3320**	**0.8958**	**0.3242**	**1.2760**	**6.7897**
42	DF-VOTE	0.8333	**0.4366**	0.2672	1.2699	0.5661	0.1694	1.0027	1.6341
	DF-WEIGHT	0.8970	0.3667	0.1774	1.2637	0.7196	0.1892	1.0862	2.0666
	SBF-BSQ	0.9221	0.3332	0.0712	1.2553	0.8510	0.2620	1.1841	4.6815
	SBF-MinVBP	0.9126	0.3976	0.0852	1.3103	0.8274	0.3124	1.2250	4.6659
	SBF-MaxVBP	**0.9587**	0.3626	**0.0372**	**1.3213**	**0.9215**	**0.3254**	**1.2841**	**9.7497**
205	DF-VOTE	0.4266	0.4328	0.4728	0.8594	-0.0462	-0.0400	0.3866	0.9155
	DF-WEIGHT	0.6865	**0.4812**	0.3586	1.1677	0.3279	0.1225	0.8091	1.3417
	SBF-BSQ	**0.9555**	0.3374	**0.0295**	**1.2929**	**0.9260**	**0.3079**	**1.2634**	**11.4373**
	SBF-MinVBP	**0.9555**	0.3374	**0.0295**	**1.2929**	**0.9260**	**0.3079**	**1.2634**	**11.4373**
	SBF-MaxVBP	**0.9555**	0.3374	**0.0295**	**1.2929**	**0.9260**	**0.3079**	**1.2634**	**11.4373**

TABLE IV

FIRST 42 BANDS USED BY SBF-BSQ, SBF-SFCTBS, AND SBF-SBCTBS

SBF-BSQ	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42
SBF-SFCTBS	11	159	138	163	154	158	135	3	88	92	84	5	155	86	137	4	105	153	28	21	36
	113	184	186	179	174	183	164	178	1	94	126	171	132	87	19	175	101	82	201	130	53
SBF-SBCTBS	186	184	164	28	183	159	158	44	132	182	171	170	4	3	2	1	5	185	181	189	160
	174	169	173	172	168	175	21	176	105	196	179	78	79	104	88	86	89	87	85	90	91

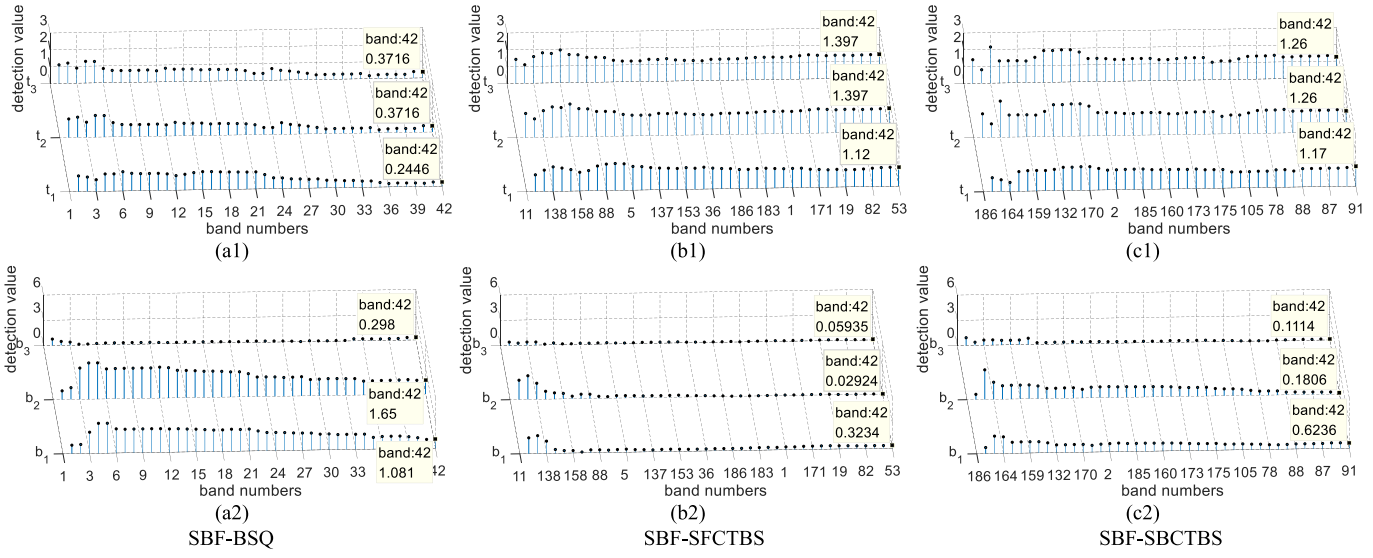

Fig. 10. Detected value fractions of three target pixels and three background pixels produced by SBF-BSQ, SBF-SFCTBS, and SBF-SBCTBS. (a1)–(a2) SBF-BSQ. (b1)–(b2) SBF-SFCTBS. (c1)–(c2) SBF-SBCTBS.

C. SBF-BP

The third experiment first sorts all bands using Minv-BP and MaxV-BP, and then fuse the band according to the sorting results for target detection. Fig. 7 shows the detected value fractions of nine target pixels and six background pixels produced by SBF-MinVBP and SBF-MaxVBP. First, as can be seen from the detection value of the last band $\mathbf{b}_{205}$, when the bands to be fused is determined, the final result of fusion is consistent with the result of direct detection using all bands. This shows that the test results are closely

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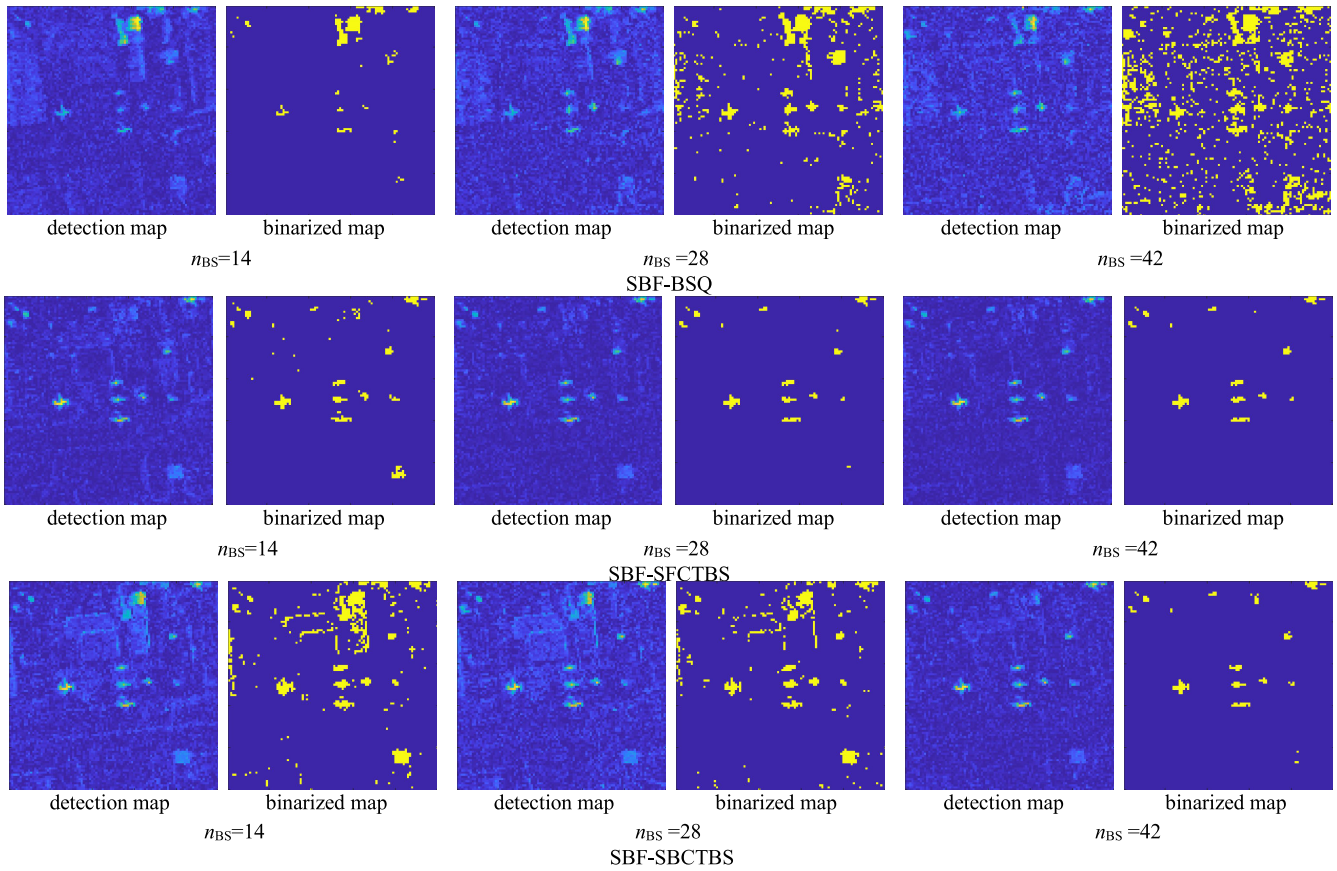

Fig. 11. Detection results and binarized image of SBF-BSQ, SBF-SFCTBS, and SBF-SBCTBS when $n_{BS} = 14$, 28, and 42.

related to which bands that are fused. Second, the adjustment of the fusion order affects the trend of the detection value. As shown in Fig. 7(b1)–(b5), the detection value produced by SBF-MaxVBP tends to be stable after only 15 bands are fused. The background suppression starts with the fusion, and the interference is also suppressed after only 15 bands are fused.

Fig. 8 shows the detection results and binarized image of SBF-BSQ, SBF-MinVBP, and SBF-MaxVBP when $n_{BS} = 14$, 28, and 42. As can be seen, no matter what n_{BS} is, the detection effect of SBF-MaxVBP is better than SBF-BSQ and SBF-MinVBP. With the increase of the number of bands, the suppression effect of SBF-MaxVBP on the background is getting better and better, while the suppression effect of SBF-BSQ is getting worse. This indicates that the detection of SBF-BSQ is still changing greatly after fusing 42 bands, and does not reach a stable state. Fig. 9 shows the 3-D ROC curves along with their generated three 2-D ROC curves of CEM detected target maps of SBF-BSQ, SBF-IBD-MinVBP, and SBF-IBD-MaxVBP and Table III calculates the AUC values. The data in bold are the best AUC produced by DF-VOTE, DF-WEIGHT, SBF-BSQ, SBF-MinVBP, and SBF-MaxVBP. The data marked with underscore indicates that the corresponding method performs better than using whole bands. The results show that the $AUC_{(D,F)}$, $AUC_{(TD)}$, $AUC_{(BS)}$, $AUC_{(TDBS)}$, $AUC_{(ODP)}$, and $AUC_{(SNPR)}$ of SBF-MaxVBP is the highest when $n_{BS} = 14$, 28, and 42. It should be noted that $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, $AUC_{(TD)}$, $AUC_{(TDBS)}$, and $AUC_{(ODP)}$ of SBF-MaxVBP are higher than that of fusing all bands when $n_{BS} = 42$. SBF-MaxVBP can effectively suppress interference of background and detect most targets by fusing a few bands, so that the detection process tends to be stable as soon as possible.

D. SBF-BS

The experiment of SBF-BP shows that a better detection result can be get with just fusing a part of bands. In this experiment, two BS methods, SFCTBS and SBCTBS, are used to select bands, and the bands are sequentially fused for target detection. Table IV listed the first 42 bands used by SBF-BSQ, SBF-SFCTBS, and SBF-SBCTBS. For comparison, Fig. 10 shows detected value fractions of three target pixels and three background pixels produced by SBF-BSQ, SBF-SFCTBS, and SBF-SBCTBS. As can be seen from Fig. 10(a1)–(a2), the detection results of SBF-BSQ in the first 42 bands are very unstable. The detection values of three target pixels t1, t2, and t3 do not increase with the increase of the band, but show a downward trend. The detection values of three background pixels b1, b2, and b3 are constantly decreasing, but the decreasing trend is very slow. In addition, when the same number of bands are fused, the detection value of some background pixels is higher than that of the target pixels, which indicates that the background seriously interferes with the detection of the target, which is undesirable. All these indicate that the detection by fusing band based

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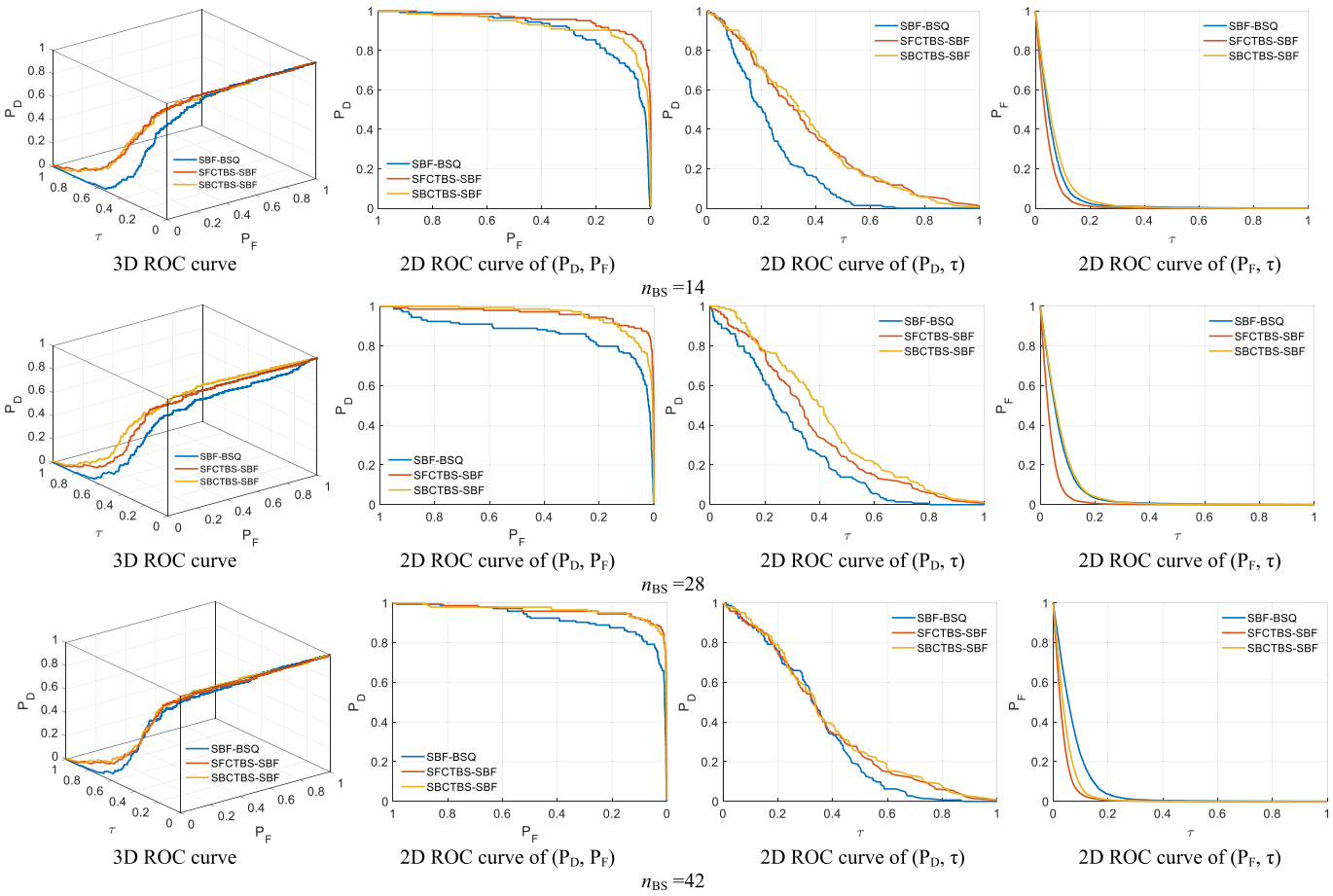

Fig. 12. 3-D ROC curves along with their generated three 2-D ROC curves of CEM detected target maps of SBF-BSQ, SBF-SFCTBS, and SBF-SBCTBS.

on SBF-BSQ is extremely unstable at the initial stage of fusion. In contrast, the detection based on SBF-SFCTBS and SBF-SBCTBS was more stable. As can be seen from Fig. 10(b1), (b2), (c1), and (c2), first, whether it is the target pixel or the background pixel, the detection value of SBF-SFCTBS and SBF-SBCTBS remains stable after only fusing six bands. Second, suppression effect on background of SBF-SFCTBS and SBF-SBCTBS is obviously better than that of SBF-BSQ. The last but not least, the detection value of the target pixel is significantly higher than the detection value of the background pixel in SBF-SFCTBS and SBF-SBCTBS, which can ensure that the target can be distinguished from the background.

To intuitively compare the detection effect of each method, Fig. 11 shows the detection results and binarized image of SBF-BSQ, SBF-SFCTBS, and SBF-SBCTBS when $n_{\mathrm{BS}} = 14$, 28, and 42. As can be seen, both SBF-SFCTBS and SBF-SBCTBS can well suppress the interference of background pixels and detect most of the target pixels when $n_{\mathrm{BS}} = 42$. The performance of SBF-SFCTBS is particularly outstanding, and most of the anomalies are detected when only 14 bands are fused. Fig. 12 shows the 3-D ROC curves along with their generated three 2-D ROC curves of CEM detected target maps of SBF-BSQ, SBF-SFCTBS, and SBF-SBCTBS, and Table V lists its corresponding AUC values. The data in bold are the best AUC produced by DF-VOTE, DF-WEIGHT, SBF-BSQ, SBF-SFCTBS, and SBF-SBCTBS. The data marked with underscore indicates that the corresponding method performs better than using whole bands. As can be seen, the $\mathrm{AUC}_{(\mathrm{D,F})}$, $\mathrm{AUC}_{(\mathrm{D},\tau)}$, $\mathrm{AUC}_{(\mathrm{TD})}$, $\mathrm{AUC}_{(\mathrm{BS})}$, $\mathrm{AUC}_{(\mathrm{TDBS})}$, $\mathrm{AUC}_{(\mathrm{ODP})}$, and $\mathrm{AUC}_{(\mathrm{SNPR})}$ of SBF-SFCTBS and SBF-SBCTBS are better than those of DF-VOTE, DF-WEIGHT, SBF-BSQ. It should be noted that $\mathrm{AUC}_{(\mathrm{D,F})}$, $\mathrm{AUC}_{(\mathrm{D},\tau)}$, $\mathrm{AUC}_{(\mathrm{TD})}$, $\mathrm{AUC}_{(\mathrm{TDBS})}$, and $\mathrm{AUC}_{(\mathrm{SNPR})}$ of SBF-SFCTBS and SBF-SBCTBS are higher than those of fusing all bands when $n_{\mathrm{BS}} = 28$ and 42. This shows that the sequence of band fusion plays an important role in the fusion result. The two fusion methods proposed in this article can not only improve the detection efficiency but also improve the detection accuracy.

E. Computational Complexity

Remote sensing data are widely used in the investigation, monitoring, analysis, prediction, and prediction of resources, environment, disasters, regions, and cities. Time performance is an important indicator to measure the quality of an algorithm. Especially in emergencies, the timeliness of data acquisition is particularly important. A good algorithm should be less time-consuming. In the traditional algorithm without SBF,

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TABLE V
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES OF DETECTION RESULTS USING DIFFERENT BAND SUBSETS

n_{BS}	Band subsets	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{ODP}	AUC_{SNPR}
14	DF-VOTE	0.6742	**0.4269**	0.3129	1.1011	0.3613	0.1140	0.7882	1.3641
	DF-WEIGHT	0.8325	0.4148	0.2050	1.2473	0.6275	0.2098	1.0423	2.0231
	SBF-BSQ	0.9015	0.2256	0.0611	1.1271	0.8404	0.1645	1.0660	3.6925
	SBF-SFCTBS	**0.9547**	**0.3613**	**0.0442**	**1.3160**	**0.9105**	**0.3171**	**1.2718**	**8.1809**
	SBF-SBCTBS	0.9241	**0.3660**	0.0685	1.2900	0.8556	0.2975	1.2216	5.3449
28	DF-VOTE	0.5443	0.4368	0.3948	0.9811	0.1495	0.0420	0.5863	1.1063
	DF-WEIGHT	0.7734	**0.4538**	0.2800	1.2272	0.4934	0.1738	0.9472	1.6207
	SBF-BSQ	0.8655	0.2789	0.0687	1.1444	0.7967	0.2102	1.0757	4.0586
	SBF-SFCTBS	**0.9610**	0.3611	**0.0366**	1.3221	**0.9244**	0.3245	1.2855	**9.8664**
	SBF-SBCTBS	0.9571	0.4134	0.0718	**1.3704**	0.8852	**0.3415**	**1.2986**	5.7543
42	DF-VOTE	0.7295	0.4078	0.2701	1.1373	0.4593	0.1377	0.8672	1.5098
	DF-WEIGHT	0.8802	**0.4130**	0.1784	1.2932	0.7018	0.2346	1.1148	2.3151
	SBF-BSQ	0.9221	0.3332	0.0712	1.2553	0.8510	0.2620	1.1841	4.6815
	SBF-SFCTBS	0.9601	0.3647	**0.0346**	1.3249	**0.9256**	0.3302	1.2903	**10.5533**
	SBF-SBCTBS	**0.9623**	0.3774	0.0452	**1.3396**	0.9171	**0.3322**	**1.2944**	8.3499
205	DF-VOTE	0.5912	**0.4193**	0.3591	1.0106	0.2321	0.0602	0.6514	1.1676
	DF-WEIGHT	0.8074	0.4117	0.2304	1.2191	0.5770	0.1814	0.9888	1.7872
	SBF-BSQ	**0.9555**	0.3374	**0.0295**	**1.2929**	**0.9260**	**0.3079**	**1.2634**	**11.4373**
	SBF-SFCTBS	**0.9555**	0.3374	**0.0295**	**1.2929**	**0.9260**	**0.3079**	**1.2634**	**11.4373**
	SBF-SBCTBS	**0.9555**	0.3374	**0.0295**	**1.2929**	**0.9260**	**0.3079**	**1.2634**	**11.4373**

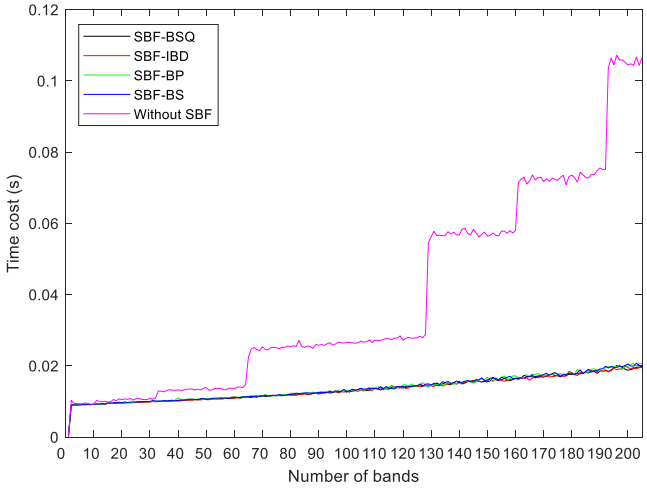

Fig. 13. Comparison of computational times among BS and various SBF methods.

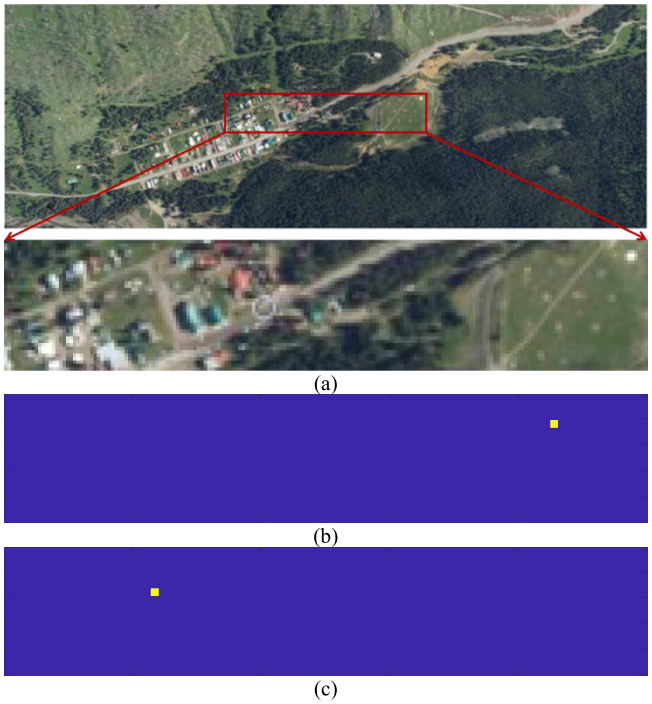

Fig. 14. (a) RGB image of HyMap data. (b), (c) Ground truth map of targets F3 and V1.

the inverse operation of large matrices is involved because the background needs to be suppressed, and the time complexity of the algorithm is $O(n^3)$. The more the number of bands, the larger the matrix to be inversed, the longer the time spent on. This is very unfavorable to the processing of hyperspectral images. The SBF proposed in this article does not involve matrix inversion in the fusion process, and the time complexity is $O(n)$, so it is suitable for processing hyperspectral images. Fig. 13 shows the comparison of computational times among BS and various SBF methods.

As can be seen, with the increase in the number of bands, the amount of data calculation is constantly increasing, leading to the time consumption of data analysis is constantly rising in BS. Due to the use of GPU to accelerate the operation of the algorithm, the upward trend is shown as a ladder. The graphics card used in the experiment is NVIDIA GeForce GTX 1050 Ti. A warp, the minimum scheduling unit in GPU, contains 32 streaming processors. Because it avoids the operation of large data, the SBF proposed in this article is not affected by the number of bands, and the data analysis time does not show an obvious upward trend with the increase of the number of bands. All the four band fusion methods can significantly shorten the analysis time.

VI. HYMAP DATA EXPERIMENTS

The second dataset was captured by the Hyperspectral Mapper (HyMap) sensor in the range of 0.45–2.5 μm, which

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TABLE VI
AUC VALUES OF DETECTION RESULTS OF F3 BY DF-VOTE, DF-WEIGHT, AND SBF-BSQ

n_{BS}	Band subsets	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{ODP}	AUC_{SNPR}
11	DF-VOTE	0.9488	0.2350	0.0988	1.1838	0.8500	0.1362	1.0850	2.3783
	DF-WEIGHT	0.9320	0.1550	**0.0749**	1.0870	0.8570	0.0801	1.0120	2.0687
	SBF-BSQ	**0.9723**	**0.2817**	0.0820	**1.2539**	**0.8903**	**0.1997**	**1.1719**	**3.4357**
22	DF-VOTE	0.8544	0.3761	0.2345	1.2306	0.6199	0.1416	0.9960	1.6036
	DF-WEIGHT	0.9919	0.3550	0.1465	1.3469	0.8454	0.2085	1.2004	2.4230
	SBF-BSQ	**0.9980**	**0.4250**	**0.0606**	**1.4230**	**0.9374**	**0.3644**	**1.3624**	**7.0145**
33	DF-VOTE	0.9660	0.5761	0.2706	1.5421	0.6954	0.3055	1.2715	2.1291
	DF-WEIGHT	0.9966	0.4994	0.1677	1.4961	0.8290	0.3318	1.3284	2.9784
	SBF-BSQ	**0.9986**	**0.5850**	**0.0692**	**1.5836**	**0.9294**	**0.5158**	**1.5144**	**8.4573**
126	DF-VOTE	0.9977	**0.7050**	0.0843	1.7027	0.9134	0.6207	1.6184	8.3625
	DF-WEIGHT	0.9972	0.6883	0.1034	1.6856	0.8938	0.5849	1.5822	6.6564
	SBF-BSQ	**0.9999**	**0.7050**	**0.0467**	**1.7049**	**0.9532**	**0.6583**	**1.6582**	**15.0896**

TABLE VII
AUC VALUES OF DETECTION RESULTS OF V1 BY DF-VOTE, DF-WEIGHT, AND SBF-BSQ

n_{BS}	Band subsets	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{ODP}	AUC_{SNPR}
11	DF-VOTE	0.8890	0.2583	0.1135	1.1473	0.7754	0.1448	1.0338	2.2756
	DF-WEIGHT	**0.9134**	0.2783	0.1204	1.1918	0.7930	0.1579	1.0713	2.3112
	SBF-BSQ	0.9105	**0.2950**	**0.1103**	**1.2055**	**0.8002**	**0.1847**	**1.0952**	**2.6745**
22	DF-VOTE	0.9328	**0.4750**	0.2408	1.4078	0.6920	0.2342	1.1670	1.9723
	DF-WEIGHT	**0.9833**	0.4328	0.1819	**1.4161**	0.8014	0.2509	1.2342	2.3794
	SBF-BSQ	0.9750	0.4317	**0.0971**	1.4067	**0.8779**	**0.3346**	**1.3096**	**4.4456**
33	DF-VOTE	0.9792	**0.6906**	0.2849	**1.6697**	0.6942	0.4056	1.3848	2.4235
	DF-WEIGHT	**0.9940**	0.5817	0.2131	1.5757	0.7809	0.3686	1.3626	2.7300
	SBF-BSQ	0.9932	0.6628	**0.1148**	1.6560	**0.8785**	**0.5480**	**1.5412**	**5.7754**
126	DF-VOTE	0.9988	0.7717	0.1036	1.7705	0.8952	0.6680	1.6669	7.4469
	DF-WEIGHT	0.9986	0.7650	0.1217	1.7636	0.8769	0.6433	1.6419	6.2866
	SBF-BSQ	**0.9999**	**0.7739**	**0.0766**	**1.7738**	**0.9233**	**0.6973**	**1.6972**	**10.1050**

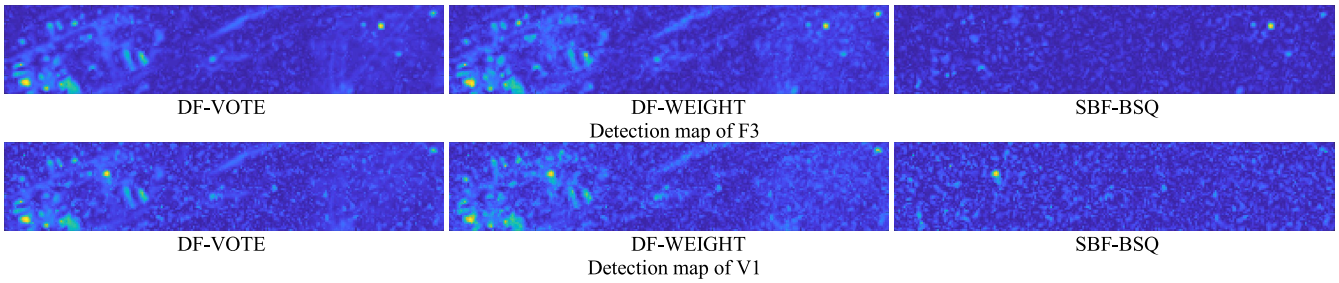

Fig. 15. Detection maps of F3 and V1 by CEM based on DF-VOTE, DF-WEIGHT, and SBF-BSQ.

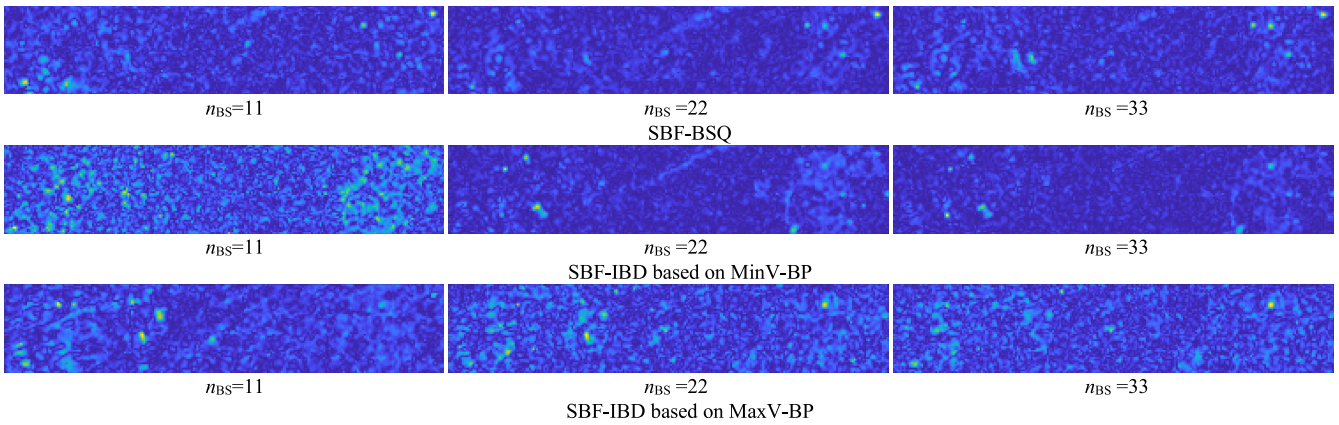

Fig. 16. Detection results of F3 by SBF-BSQ, SBF-IBD based on MinV-BP, and MaxV-BP when $n_{BS} = 11$, 22, and 33.

is free and available online [36], [37]. As shown in Fig. 14, the data provided by RIT university was collected around the Cooke City, Montanta, on July 4, 2006. The image is in the size of 280 × 400 pixels, with 126 bands. To improve the computational efficiency, we intercepted part of the images (50 × 250 × 126) for experiments. Several test fabric

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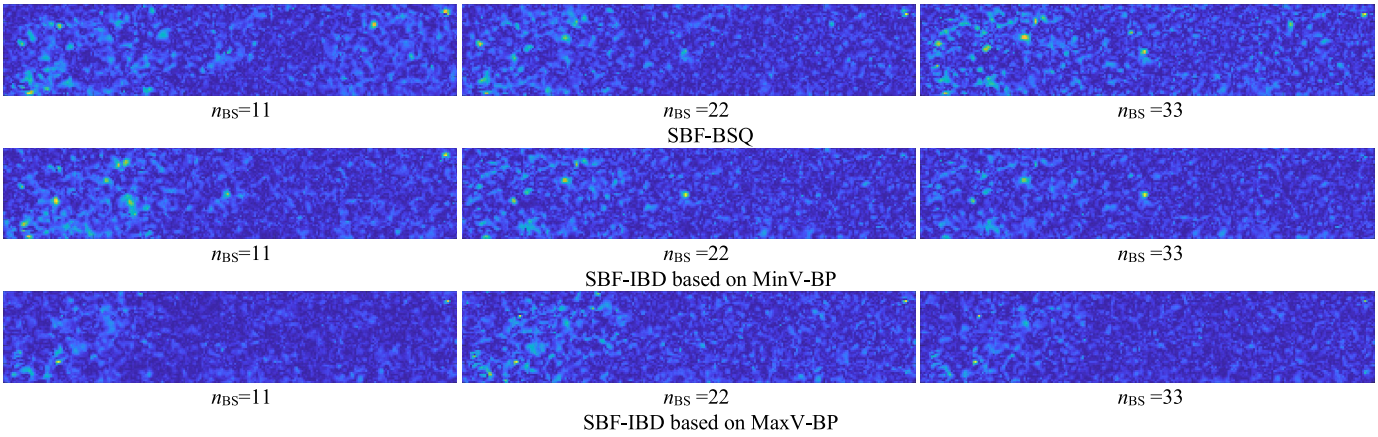

Fig. 17. Detection results of V1 by SBF-BSQ, SBF-IBD based on MinV-BP, and MaxV-BP when $n_{BS} = 11$, 22, and 33.

TABLE VIII
AUC VALUES OF DETECTION RESULTS OF F3 USING DIFFERENT BAND SUBSETS

n_{BS}	Band subsets	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{ODP}	AUC_{SNPR}
11	DF-VOTE	0.8265	0.4106	0.2748	1.2370	0.5517	0.1358	0.9622	1.4941
	DF-WEIGHT	0.9654	0.4694	0.2313	1.4349	0.7342	0.2382	1.2036	2.0300
	SBF-BSQ	0.9723	0.2817	**0.0820**	1.2539	**0.8903**	0.1997	1.1719	**3.4357**
	SBF-IBD-MinVBP	**0.9786**	**0.5339**	0.1635	**1.5125**	0.8151	**0.3704**	**1.3490**	3.2657
	SBF-IBD-MaxVBP	0.9714	0.3472	0.1039	1.3187	0.8675	0.2433	1.2147	3.3410
22	DF-VOTE	0.9847	0.7072	0.3024	1.6920	0.6823	0.4048	1.3895	2.3385
	DF-WEIGHT	0.9963	**0.7494**	0.2684	**1.7457**	0.7278	0.4810	1.4773	2.7918
	SBF-BSQ	0.9980	0.4250	**0.0606**	1.4230	**0.9374**	0.3644	1.3624	**7.0145**
	SBF-IBD-MinVBP	0.9957	0.4006	0.0620	1.3963	0.9337	0.3386	1.3343	6.4601
	SBF-IBD-MaxVBP	**0.9985**	0.7006	0.1215	1.6991	0.8770	**0.5790**	**1.5775**	5.7650
33	DF-VOTE	0.9950	0.7583	0.2716	1.7533	0.7234	0.4867	1.4817	2.7916
	DF-WEIGHT	0.9988	**0.8194**	0.2515	**1.8183**	0.7473	0.5679	1.5667	3.2577
	SBF-BSQ	0.9986	0.5850	0.0692	1.5836	0.9294	0.5158	1.5144	**8.4573**
	SBF-IBD-MinVBP	0.9975	0.4228	**0.0528**	1.4203	**0.9447**	0.3700	1.3675	8.0035
	SBF-IBD-MaxVBP	**0.9992**	0.7572	0.1143	1.7565	0.8849	**0.6429**	**1.6421**	6.6232

TABLE IX
AUC VALUES OF DETECTION RESULTS OF V1 USING DIFFERENT BAND SUBSETS

n_{BS}	Band subsets	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{ODP}	AUC_{SNPR}
11	DF-VOTE	0.8836	0.3639	0.2258	1.2475	0.6578	0.1381	1.0216	1.6115
	DF-WEIGHT	0.9739	**0.3872**	0.1769	**1.3611**	0.7970	0.2103	1.1842	2.1890
	SBF-BSQ	0.9105	0.2950	0.1103	1.2055	0.8002	0.1847	1.0952	2.6745
	SBF-IBD-MinVBP	0.8742	0.3572	0.1044	1.2314	0.7698	**0.2528**	1.1270	3.4209
	SBF-IBD-MaxVBP	**0.9860**	0.2883	**0.0776**	1.2743	**0.9084**	0.2107	**1.1967**	**3.7144**
22	DF-VOTE	0.9691	0.4117	0.1380	1.3808	0.8312	0.2737	1.2428	2.9841
	DF-WEIGHT	0.9691	0.3817	0.1301	1.3507	0.8389	0.2515	1.2206	2.9327
	SBF-BSQ	0.9750	0.4317	**0.0971**	1.4067	0.8779	0.3346	1.3096	4.4456
	SBF-IBD-MinVBP	**0.9890**	**0.5061**	0.0981	**1.4951**	**0.8909**	**0.4080**	**1.3971**	**5.1609**
	SBF-IBD-MaxVBP	0.9878	0.4139	0.1014	1.4017	0.8864	0.3125	1.3003	4.0822
33	DF-VOTE	0.9687	0.4061	0.1581	1.3748	0.8107	0.2480	1.2168	2.5690
	DF-WEIGHT	0.9798	0.4350	0.1516	1.4148	0.8282	0.2834	1.2632	2.8687
	SBF-BSQ	0.9932	**0.6628**	0.1148	**1.6560**	0.8785	**0.5480**	**1.5412**	**5.7754**
	SBF-IBD-MinVBP	0.9959	0.5072	0.0910	1.5031	0.9049	0.4163	1.4121	5.5768
	SBF-IBD-MaxVBP	**0.9966**	0.3983	**0.0791**	1.3950	**0.9175**	0.3192	1.3159	5.0354

panels (F1–F4) and vehicles (V1–V3) were deployed, and we randomly choose two of them (F3 and V1) to corroborate the theory of SBF. Fig. 1(b) and (c) shows the ground truth images of targets F3 and V1, respectively. In this section, we use the HyMap data in Fig. 14 to repeat the experiment in Section IV.

A. Sequential BF According to BSQ

The fabric panel F3 and vehicles V1 were regarded as the interested targets. Analogous to the experiment of the airport scene, the CEM detector was used to detect F3 and V1 in the HyMap data based on DF-VOTE, DF-WEIGHT,

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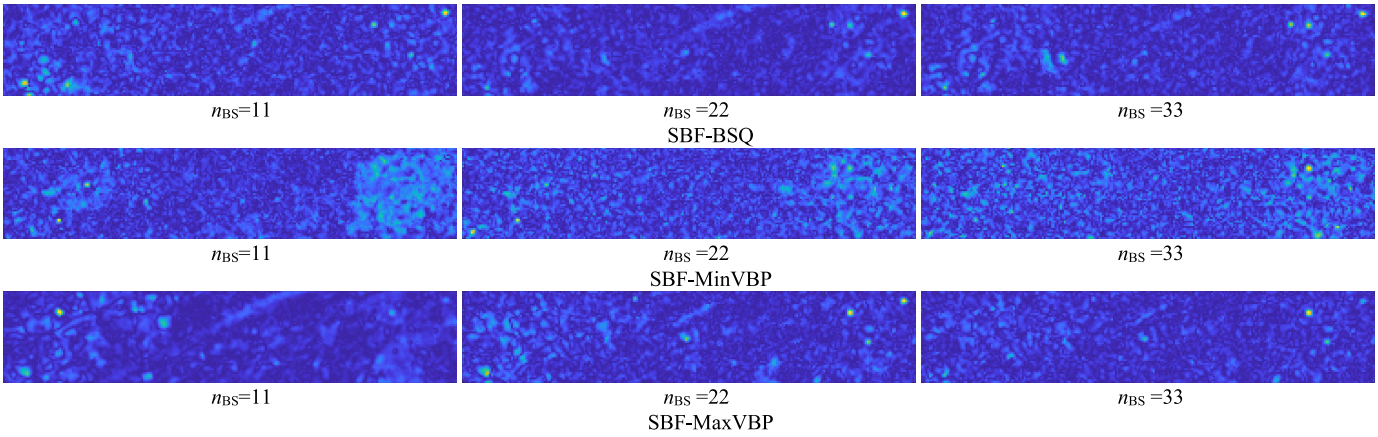

Fig. 18. Detection results of F3 by SBF-BSQ, SBF-MinVBP, and SBF-MaxVBP when $n_{BS} = 11$, 22, and 33.

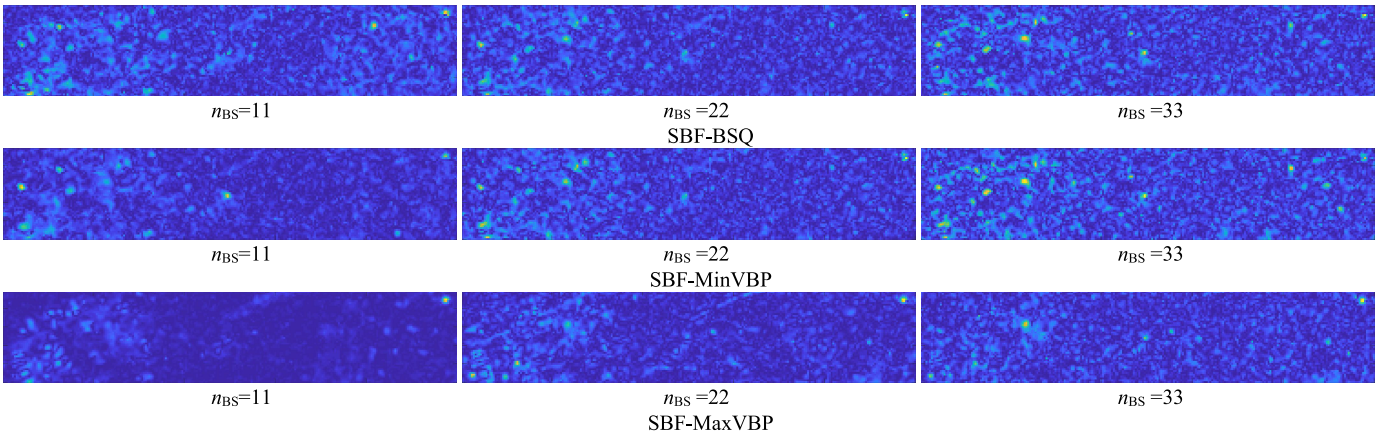

Fig. 19. Detection results of V1 by SBF-BSQ, SBF-MinVBP, and SBF-MaxVBP when $n_{BS} = 11$, 22, and 33.

TABLE X
AUC VALUES OF DETECTION RESULTS OF F3 USING DIFFERENT BAND SUBSETS

n_{BS}	Band subsets	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{ODP}	AUC_{SNPR}
11	DF-VOTE	0.8267	0.4194	0.2750	1.2462	0.5518	0.1445	0.9712	1.5255
	DF-WEIGHT	0.9776	**0.4994**	0.2292	**1.4770**	0.7483	**0.2702**	1.2478	2.1787
	SBF-BSQ	0.9723	0.2817	0.0820	1.2539	0.8903	0.1997	1.1719	3.4357
	SBF-MinVBP	0.9755	0.3783	0.1148	1.3538	0.8607	0.2636	1.2390	3.2963
	SBF-MaxVBP	**0.9907**	0.3217	**0.0624**	1.3124	**0.9283**	0.2593	**1.2500**	**5.1586**
22	DF-VOTE	0.9977	0.6872	0.0881	1.6849	0.9096	0.5991	1.5968	7.8020
	DF-WEIGHT	0.9948	0.3750	0.0614	1.3698	0.9334	0.3136	1.3084	6.1046
	SBF-BSQ	0.9980	0.4250	**0.0606**	1.4230	**0.9374**	0.3644	1.3624	7.0145
	SBF-MinVBP	0.9968	0.5239	0.1153	1.5207	0.8815	0.4086	1.4054	4.5454
	SBF-MaxVBP	**0.9991**	**0.7050**	0.0749	**1.7041**	0.9242	**0.6301**	**1.6292**	**9.4092**
33	DF-VOTE	0.9865	0.7072	0.2664	1.6937	0.7201	0.4408	1.4274	2.6549
	DF-WEIGHT	0.9989	**0.7706**	0.2212	**1.7694**	0.7777	0.5494	1.5482	3.4836
	SBF-BSQ	0.9986	0.5850	0.0692	1.5836	0.9294	0.5158	1.5144	8.4573
	SBF-MinVBP	0.9994	0.7328	0.1291	1.7321	0.8702	0.6036	1.6030	5.6746
	SBF-MaxVBP	**0.9998**	0.7117	**0.0652**	1.7115	**0.9346**	**0.6465**	**1.6463**	**10.9131**

and SBF-BSQ. The n_{BS} of HyMap data is calculated to be 11 by VD algorithm [10, Chapter 4], [28]–[31]. Fig. 15 shows the detection map of F3 and V1 by CEM based on DF-VOTE, DF-WEIGHT, and SBF-BSQ. As can be seen, all three fusion methods can accurately detect the target. However, the suppression effect of DF-VOTE and DF-WEIGHT on the background is obviously not as good as that of SBF-BSQ. No matter in the detection of F3 or V1, there are a large number of background pixels and uninterested target pixels in the detection result of decision fusion. In contrast, the SBF-BSQ can greatly reduce the interference of the uninterested target pixels, and effectively suppress interference of the background while accurately detecting the target cell.

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TABLE XI
AUC VALUES OF DETECTION RESULTS OF V1 USING DIFFERENT BAND SUBSETS

n_{BS}	Band subsets	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{ODP}	AUC_{SNPR}
11	DF-VOTE	0.8597	0.1739	0.0979	1.0336	0.7618	0.0760	0.9356	1.7755
	DF-WEIGHT	0.9007	0.1972	0.0923	1.0979	0.8084	0.1049	1.0056	2.1369
	SBF-BSQ	0.9105	**0.2950**	0.1103	**1.2055**	0.8002	**0.1847**	1.0952	2.6745
	SBF-MinVBP	0.9377	0.2606	0.0770	1.1983	0.8607	0.1835	**1.1212**	3.3821
	SBF-MaxVBP	**0.9574**	0.1539	**0.0369**	1.1113 e	**0.9205**	0.1170	1.0744	**4.1686**
22	DF-VOTE	0.9027	0.4039	0.2150	1.3066	0.6877	0.1889	1.0915	1.8784
	DF-WEIGHT	0.9864	0.3494	0.1377	1.3359	0.8488	0.2118	1.1982	2.5378
	SBF-BSQ	0.9750	0.4317	0.0971	1.4067	0.8779	0.3346	1.3096	4.4456
	SBF-MinVBP	0.9684	**0.4428**	0.0998	1.4111	0.8685	0.3430	1.3113	4.4358
	SBF-MaxVBP	**0.9957**	0.4372	**0.0728**	**1.4329**	**0.9229**	**0.3644**	**1.3601**	**6.0041**
33	DF-VOTE	0.9973	0.6094	0.1102	1.6068	0.8872	0.4993	1.4966	5.5322
	DF-WEIGHT	0.9966	0.4139	**0.0872**	1.4105	0.9094	0.3267	1.3233	4.7482
	SBF-BSQ	0.9932	0.6628	0.1148	1.6560	0.8785	0.5480	1.5412	5.7754
	SBF-MinVBP	0.9884	**0.7017**	0.1251	**1.6901**	0.8633	**0.5766**	1.5650	5.6088
	SBF-MaxVBP	**0.9991**	0.6583	0.0875	1.6574	**0.9116**	0.5708	**1.5699**	**7.5252**

TABLE XII
FIRST 33 BANDS USED BY SBF-BSQ, SBF-SFCTBS, AND SBF-SBCTBS

SBF-BSQ	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
SBF-SFCTBS	83	18	1	9	126	2	5	82	64	63	38	19	65	10	32	35	26
	21	24	111	105	66	124	123	36	94	112	108	109	20	85	90	91	
SBF-SBCTBS	21	2	108	82	3	74	38	111	99	90	24	42	124	116	13	14	55
	63	80	35	1	4	5	6	7	33	75	92	100	19	22	56	25	

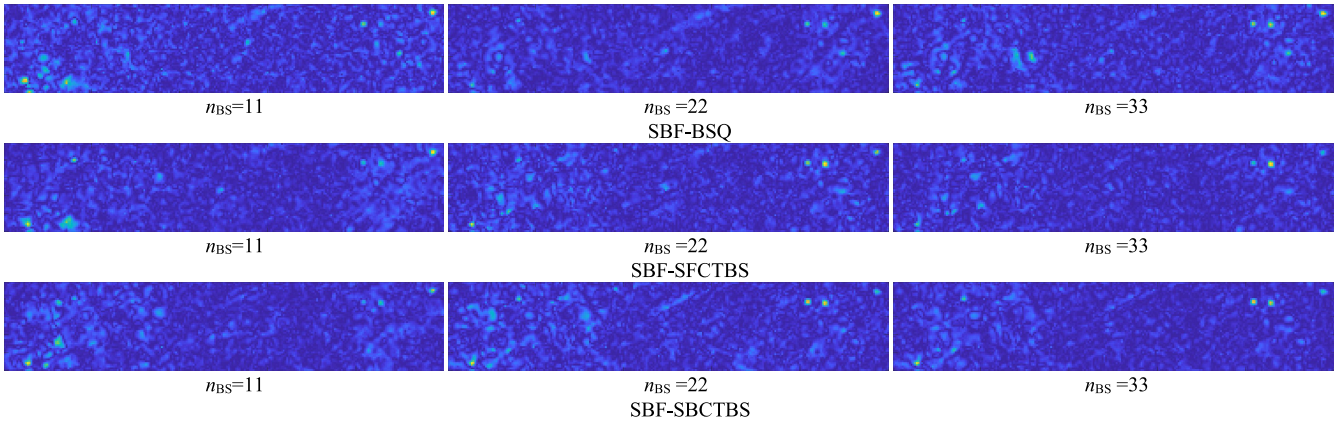

Fig. 20. Detection results of F3 by SBF-BSQ, SBF-SFCTBS, and SBF-SBCTBS when $n_{BS} = 11$, 22, and 33.

To more accurately evaluate the detection effects of various fusion methods, Tables VI and VII list the AUC values of detection results of F3 and V1 by DF-VOTE, DF-WEIGHT, and SBF-BSQ. The data in bold are the best AUC produced by DF-VOTE, DF-WEIGHT, and SBF-BSQ. As can be seen, almost all of the best AUC values in detecting F3 are obtained in SBF-BSQ. In the detection of V1, both DF-WEIGHT and SBF-BSQ can achieve a good detection effect. It is worth noting that $AUC_{(F,\tau)}$, $AUC_{(BS)}$, $AUC_{(TDBS)}$, $AUC_{(ODP)}$, and $AUC_{(SNPR)}$ of SBF-BSQ are higher than that of DF-VOTE and DF-WEIGHT when using different number of bands.

B. SBF-IBD

In this section, SBF-IBD was performed on the HyMap data. Band 21 and band 76 were selected as the highest priority bands of the HyMap data by (20) and (21). Then, band 21 and band 76 were to be fused as the initial band. Figs. 16 and 17 show the detection results of F3 and V1 by SBF-BSQ, SBF-IBD based on MinV-BP and MaxV-BP when $n_{BS} = 11$, 22, and 33. These results only provided preliminary analysis. To conduct detailed quantitative studies, Tables VIII and IX list the AUC values of detection results of F3 and V1 using different band subsets. The data in bold are the best AUC produced by DF-VOTE, DF-WEIGHT, SBF-BSQ, SBF-IBD-MinVBP, and SBF-IBD-MaxVBP. As can be seen, the results show that the fusion effect of three SBF methods is better than that of two DF methods. In three SBF methods, the detection effect of SBF-IBD is better than SBF-BSQ.

C. SBF-BP

The third experiment performing SBF-BP on the HyMap data. All bands were sorted using MinV-BP and MaxV-BP,

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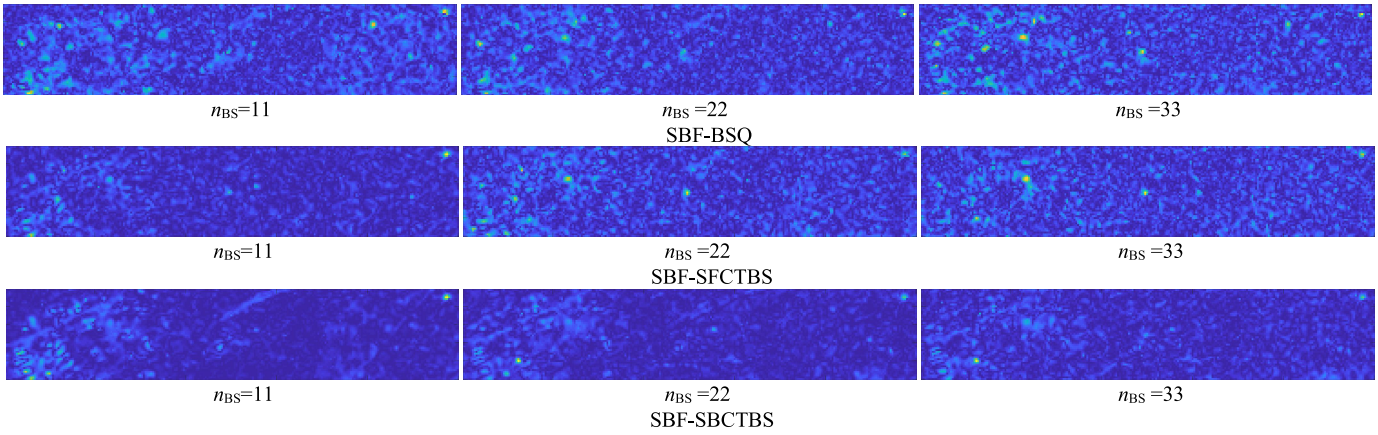

Fig. 21. Detection results of V1 by SBF-BSQ, SBF-SFCTBS, and SBF-SBCTBS when $n_{BS} = 11$, 22, and 33.

TABLE XIII
AUC VALUES OF DETECTION RESULTS OF F3 USING DIFFERENT BAND SUBSETS

n_{BS}	Band subsets	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{ODP}	AUC_{SNPR}
11	DF-VOTE	0.8380	0.3850	0.2468	1.2230	0.5912	0.1382	0.9762	1.5597
	DF-WEIGHT	0.9870	0.3639	0.1570	1.3509	0.8300	0.2069	1.1939	2.3180
	SBF-BSQ	0.9723	0.2817	0.0820	1.2539	0.8903	0.1997	1.1719	3.4357
	SBF-SFCTBS	0.9946	**0.4094**	0.0677	**1.4040**	0.9269	**0.3417**	**1.3363**	**6.0464**
	SBF-SBCTBS	**0.9947**	0.3906	**0.0674**	1.3852	**0.9272**	0.3231	1.3178	5.7928
22	DF-VOTE	0.9963	0.6017	0.1000	1.5980	0.8963	0.5017	1.4980	6.0186
	DF-WEIGHT	0.9922	0.3717	0.0784	1.3639	0.9138	0.2933	1.2855	4.7414
	SBF-BSQ	0.9980	0.4250	**0.0606**	1.4230	**0.9374**	0.3644	1.3624	7.0145
	SBF-SFCTBS	**0.9994**	**0.7028**	0.0674	**1.7021**	0.9320	**0.6354**	**1.6347**	**10.4250**
	SBF-SBCTBS	0.9989	0.6317	0.0712	1.6306	0.9278	0.5605	1.5594	8.8735
33	DF-VOTE	0.9535	0.5694	0.2554	1.5229	0.6981	0.3140	1.2675	2.2296
	DF-WEIGHT	0.9961	0.5117	0.1688	1.5078	0.8273	0.3428	1.3390	3.0306
	SBF-BSQ	0.9986	0.5850	0.0692	1.5836	0.9294	0.5158	1.5144	8.4573
	SBF-SFCTBS	**0.9998**	**0.7139**	**0.0572**	**1.7137**	**0.9426**	**0.6567**	**1.6564**	**12.4727**
	SBF-SBCTBS	0.9988	0.5694	0.0574	1.5683	0.9414	0.5121	1.5109	9.9222

TABLE XIV
AUC VALUES OF DETECTION RESULTS OF V1 USING DIFFERENT BAND SUBSETS

n_{BS}	Band subsets	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{ODP}	AUC_{SNPR}
11	DF-VOTE	0.6707	**0.3561**	0.2689	1.0268	0.4017	0.0872	0.7578	1.3242
	DF-WEIGHT	0.9388	0.2806	0.1603	1.2194	0.7786	0.1203	1.0591	1.7506
	SBF-BSQ	0.9105	0.2950	0.1103	1.2055	0.8002	0.1847	1.0952	2.6745
	SBF-SFCTBS	0.9483	0.2839	0.0623	**1.2322**	0.8860	**0.2216**	**1.1699**	**4.5566**
	SBF-SBCTBS	**0.9543**	0.2028	**0.0507**	1.1571	**0.9036**	0.1521	1.1064	3.9993
22	DF-VOTE	0.9017	0.3794	0.2077	1.2811	0.6940	0.1718	1.0735	1.8272
	DF-WEIGHT	0.9779	0.3350	0.1367	1.3129	0.8412	0.1983	1.1762	2.4503
	SBF-BSQ	0.9750	0.4317	0.0971	1.4067	0.8779	0.3346	1.3096	4.4456
	SBF-SFCTBS	**0.9965**	**0.5883**	0.0992	**1.5848**	0.8973	**0.4891**	**1.4856**	**5.9292**
	SBF-SBCTBS	0.9916	0.2683	**0.0509**	1.2599	**0.9407**	0.2175	1.2090	5.2740
33	DF-VOTE	0.8458	0.4139	0.2645	1.2597	0.5813	0.1494	0.9952	1.5648
	DF-WEIGHT	0.9866	0.4439	0.1933	1.4305	0.7933	0.2506	1.2371	2.2962
	SBF-BSQ	0.9932	0.6628	0.1148	1.6560	0.8785	0.5480	1.5412	5.7754
	SBF-SFCTBS	**0.9992**	**0.7372**	0.1004	**1.7364**	0.8988	**0.6368**	**1.6360**	**7.3415**
	SBF-SBCTBS	0.9975	0.3683	**0.0605**	1.3658	**0.9369**	0.3078	1.3053	6.0871

and then fuse the band according to the sorting results for target detection. Figs. 18 and 19 show the detection results of F3 and V1 by SBF-BSQ, SBF-MinVBP, and SBF-MaxVBP when $n_{BS} = 11$, 22, and 33. As can be seen, no matter what n_{BS} is, the suppression effect of SBF-MaxVBP on the background is significantly better than that of SBF-BSQ and SBF-MinVBP. Tables X and XI calculate the AUC values of detection results of F3 and V1 using different band subsets. The data in bold are the best AUC produced by DF-VOTE, DF-WEIGHT, SBF-BSQ, SBF-MinVBP, and SBF-MaxVBP. The results show that the $AUC_{(D,F)}$, $AUC_{(ODP)}$, and $AUC_{(SNPR)}$ of SBF-MaxVBP is the highest when $n_{BS} = 11$, 22, and 33.

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SBF-MaxVBP can effectively suppress interference of background and detect targets by fusing a few bands.

D. SBF-BS

The fourth experiment performing SBF-BS on the HyMap data. SFCTBS and SBCTBS are used to select bands, and the bands selected are sequentially fused for target detection. Table XII lists the first 33 bands used by SBF-BSQ, SBF-SFCTBS, and SBF-SBCTBS. To intuitively compare the detection effect of each method, Figs. 20 and 21 show the detection results of F3 and V1 by SBF-BSQ, SBF-SFCTBS, and SBF-SBCTBS when $n_{\mathrm{BS}} = 11$, 22, and 33, and Tables XIII and XIV list their corresponding AUC values. The data in bold are the best AUC produced by DF-VOTE, DF-WEIGHT, SBF-BSQ, SBF-SFCTBS, and SBF-SBCTBS. As can be seen, both SBF-SFCTBS and SBF-SBCTBS can well suppress the interference of background pixels and detect targets using fewer bands. This shows that the sequence of fusion is an important factor. The two fusion methods proposed in this article can improve the efficiency and accuracy of detection.

VII. Conclusion

This article proposed an SBF method for hyperspectral target detection and gave a detailed derivation of the fusion theory. Four versions of SBF, namely SBF-BSQ, SBF-IBD, SBF-BP, and SBF-BS, are introduced for application. These four methods can be used in different application scenarios according to different requirements and can improve the application effect step by step. SBF-BSQ can be used for real-time target detection. Unlike the target detection based on BS which needs to wait for all the bands to be received before starting the detection, SBF-BSQ can detect the target while receiving the band because the fusion is based on the sensor acquisition sequence BSQ. Different from SBF-BSQ, SBF-IBD can design different initial bands for different applications to improve

$$\begin{aligned}
\delta^{\mathrm{CEM}}_{\mathbf{b}_s\mathbf{b}_t} &= k^{-1}_{\mathbf{b}_s\mathbf{b}_t}\mathbf{d}^T_{\mathbf{b}_s\mathbf{b}_t}\mathbf{R}^{-1}_{\mathbf{b}_s\mathbf{b}_t}\mathbf{r}_{\mathbf{b}_s\mathbf{b}_t} = k^{-1}_{\mathbf{b}_s\mathbf{b}_t}(d_{\mathbf{b}_s}, d_{\mathbf{b}_t})\begin{bmatrix}\mathbf{R}^{-1}_{\mathbf{b}_s} + \frac{1}{N}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} & -\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \\ -\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} & N\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\end{bmatrix}\begin{pmatrix} r_{\mathbf{b}_s} \\ r_{\mathbf{b}_t}\end{pmatrix} \\
&= k^{-1}_{\mathbf{b}_s\mathbf{b}_t}d_{\mathbf{b}_s}\mathbf{R}^{-1}_{\mathbf{b}_s}r_{\mathbf{b}_s} + \frac{1}{N}k^{-1}_{\mathbf{b}_s\mathbf{b}_t}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}d_{\mathbf{b}_s}\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}r_{\mathbf{b}_s} - k^{-1}_{\mathbf{b}_s\mathbf{b}_t}d_{\mathbf{b}_t}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}r_{\mathbf{b}_s} - k^{-1}_{\mathbf{b}_s\mathbf{b}_t}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}d_{\mathbf{b}_s}\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}r_{\mathbf{b}_t} \\
&\quad + Nk^{-1}_{\mathbf{b}_s\mathbf{b}_t}d_{\mathbf{b}_t}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}r_{\mathbf{b}_t} \\
&= k^{-1}_{\mathbf{b}_s\mathbf{b}_t}k_{\mathbf{b}_s}\delta^{\mathrm{CEM}}_{\mathbf{b}_s} + \frac{1}{N}k^{-1}_{\mathbf{b}_s\mathbf{b}_t}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}(d_{\mathbf{b}_s}\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} - Nd_{\mathbf{b}_t})\left(\upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}r_{\mathbf{b}_s} - Nr_{\mathbf{b}_t}\right) \qquad \text{(A1)}
\end{aligned}$$

$$\begin{aligned}
k_{\mathbf{b}_s\mathbf{b}_t} &= \mathbf{d}^T_{\mathbf{b}_s\mathbf{b}_t}\mathbf{R}^{-1}_{\mathbf{b}_s\mathbf{b}_t}\mathbf{d}_{\mathbf{b}_s\mathbf{b}_t} = (d_{\mathbf{b}_s}, d_{\mathbf{b}_t})\begin{bmatrix}\mathbf{R}^{-1}_{\mathbf{b}_s} + \frac{1}{N}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} & -\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} \\ -\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} & N\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\end{bmatrix}\begin{pmatrix} d_{\mathbf{b}_s} \\ d_{\mathbf{b}_t}\end{pmatrix} \\
&= d_{\mathbf{b}_s}\mathbf{R}^{-1}_{\mathbf{b}_s}d_{\mathbf{b}_s} + \frac{1}{N}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}d_{\mathbf{b}_s}\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\upsilon^T_{\mathbf{b}_s\mathbf{b}_t\mathbf{b}_s}d_{\mathbf{b}_s} - d_{\mathbf{b}_t}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}\upsilon^T_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}d_{\mathbf{b}_s} - \beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}d_{\mathbf{b}_s}\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}d_{\mathbf{b}_t} \\
&\quad + Nd_{\mathbf{b}_t}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}d_{\mathbf{b}_t} \\
&= k_{\mathbf{b}_s} + \frac{1}{N}\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s}(d_{\mathbf{b}_s}\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} - Nd_{\mathbf{b}_t})^2 \qquad \text{(A2)}
\end{aligned}$$

$$\begin{aligned}
\delta^{\mathrm{CEM}}_{\mathbf{B}_{s,m}\mathbf{b}_t} &= k^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}\mathbf{d}^T_{\mathbf{B}_{s,m}\mathbf{b}_t}\mathbf{R}^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}\mathbf{r}_{\mathbf{B}_{s,m}\mathbf{b}_t} = k^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}\left(d^T_{\mathbf{B}_{s,m}}, d_{\mathbf{b}_t}\right) \\
&\quad \times \begin{bmatrix}\mathbf{R}^{-1}_{\mathbf{B}_{s,m}} + \frac{1}{N}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} & -\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \\ -\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} & N\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\end{bmatrix}\begin{pmatrix}\mathbf{r}_{\mathbf{B}_{s,m}} \\ \mathbf{r}_{\mathbf{b}_t}\end{pmatrix} \\
&= k^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}d^T_{\mathbf{B}_{s,m}}\mathbf{R}^{-1}_{\mathbf{B}_{s,m}}\mathbf{r}_{\mathbf{B}_{s,m}} + \frac{1}{N}k^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}d^T_{\mathbf{B}_{s,m}}\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\mathbf{r}_{\mathbf{B}_{s,m}} - k^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}d_{\mathbf{b}_t}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\mathbf{r}_{\mathbf{B}_{s,m}} \\
&\quad - k^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}d^T_{\mathbf{B}_{s,m}}\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\mathbf{r}_{\mathbf{b}_t} + Nk^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}d_{\mathbf{b}_t}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\mathbf{r}_{\mathbf{b}_t} \\
&= k^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}k_{\mathbf{B}_{s,m}}\delta^{\mathrm{CEM}}_{\mathbf{B}_{s,m}} + \frac{1}{N}k^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\left(d^T_{\mathbf{B}_{s,m}}\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} - Nd_{\mathbf{b}_t}\right)\left(\upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\mathbf{r}_{\mathbf{B}_{s,m}} - N\mathbf{r}_{\mathbf{b}_t}\right) \qquad \text{(B1)}
\end{aligned}$$

$$\begin{aligned}
k_{\mathbf{B}_{s,m}\mathbf{b}_t} &= d^T_{\mathbf{B}_{s,m}\mathbf{b}_t}\mathbf{R}^{-1}_{\mathbf{B}_{s,m}\mathbf{b}_t}d_{\mathbf{B}_{s,m}\mathbf{b}_t} = \left(d^T_{\mathbf{B}_{s,m}}, d_{\mathbf{b}_t}\right)\begin{bmatrix}\mathbf{R}^{-1}_{\mathbf{B}_{s,m}} + \frac{1}{N}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} & -\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} \\ -\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} & N\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\end{bmatrix}\begin{pmatrix} d_{\mathbf{B}_{s,m}} \\ d_{\mathbf{b}_t}\end{pmatrix} \\
&= d^T_{\mathbf{B}_{s,m}}\mathbf{R}^{-1}_{\mathbf{B}_{s,m}}d_{\mathbf{B}_{s,m}} + \frac{1}{N}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}d^T_{\mathbf{B}_{s,m}}\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}d_{\mathbf{B}_{s,m}} - d_{\mathbf{b}_t}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\upsilon^T_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}d_{\mathbf{B}_{s,m}} \\
&\quad - \beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}d^T_{\mathbf{B}_{s,m}}\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}d_{\mathbf{b}_t} + Nd_{\mathbf{b}_t}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}d_{\mathbf{b}_t} \\
&= k_{\mathbf{B}_{s,m}} + \frac{1}{N}\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}}\left(d^T_{\mathbf{B}_{s,m}}\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} - Nd_{\mathbf{b}_t}\right)^2 \qquad \text{(B2)}
\end{aligned}$$

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the initial detection effect and thus improve the detection efficiency. Not satisfied with the improvement of the initial effect by SBF-IBD, SBF-BP can design unique fusion orders for different requirements and can fuse fewer bands to make the detection process stable. As can be seen from the experimental result of SBF-BP, the target can be detected without fusing all bands. Therefore, SBF-BS is designed to obtain better detection results by fusing some bands, which greatly accelerate the detection speed. All four methods improve the accuracy and efficiency of target detection at different levels.

The experimental results prove the advantages of SBF in target detection. SBF methods can obtain higher detection accuracy by fusing fewer bands. The methods proposed in this article avoid some problems existing in BS. The advantages of SBF can be summarized as follows.

First, SBF is not necessary to design strategies to search for the optimal solution and does not need to determine n_{BS} in advance. This reduces the computation of data processing and improves detection efficiency.

Second, in the detection based on BS, the set of selected bands is immutable once the detection is started. In the detection based on band fusion, the fusion and verification are run in parallel. The order of the bands can be dynamically modulated according to the application effect during the whole process of fusion, which improves the flexibility of the detection process.

Third, SBF can provide the detailed change trend of application effect, so as to judge the influence of each band on the application effect according to the change trend.

Finally, data transmission, band fusion, and target detection are performed simultaneously in SBF. It not only improves the speed of data processing but also reduces the requirement of hardware configuration, which is of great significance to the data processing on the satellite.

APPENDIX A

See (A1) and (A2), as shown at the bottom of the previous page, where $\upsilon_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} = \mathbf{R}_{\mathbf{b}_s}^{-1}\mathbf{x}_{\mathbf{b}_s}\mathbf{x}_{\mathbf{b}_t}$, and $\beta_{\mathbf{b}_s\mathbf{b}_t|\mathbf{b}_s} = (\mathbf{x}_{\mathbf{b}_t}^T[P_{\mathbf{X}_{\mathbf{b}_s}^{\mathrm{T}}}^{\perp}]\mathbf{x}_{\mathbf{b}_t})^{-1}$.

APPENDIX B

See (B1) and (B2), as shown at the bottom of the previous page, where $\upsilon_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} = \mathbf{R}_{\mathbf{B}_{s,m}}^{-1}\mathbf{X}_{\mathbf{B}_{s,m}}\mathbf{x}_{\mathbf{b}_t}$, and $\beta_{\mathbf{B}_{s,m}\mathbf{b}_t|\mathbf{B}_{s,m}} = (\mathbf{x}_{\mathbf{b}_t}^T[P_{\mathbf{X}_{\mathbf{B}_{s,m}}^{\mathrm{T}}}^{\perp}]\mathbf{x}_{\mathbf{b}_t})^{-1}$.

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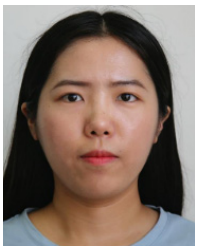

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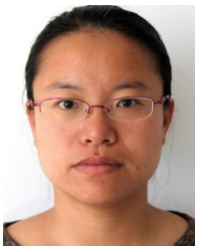

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